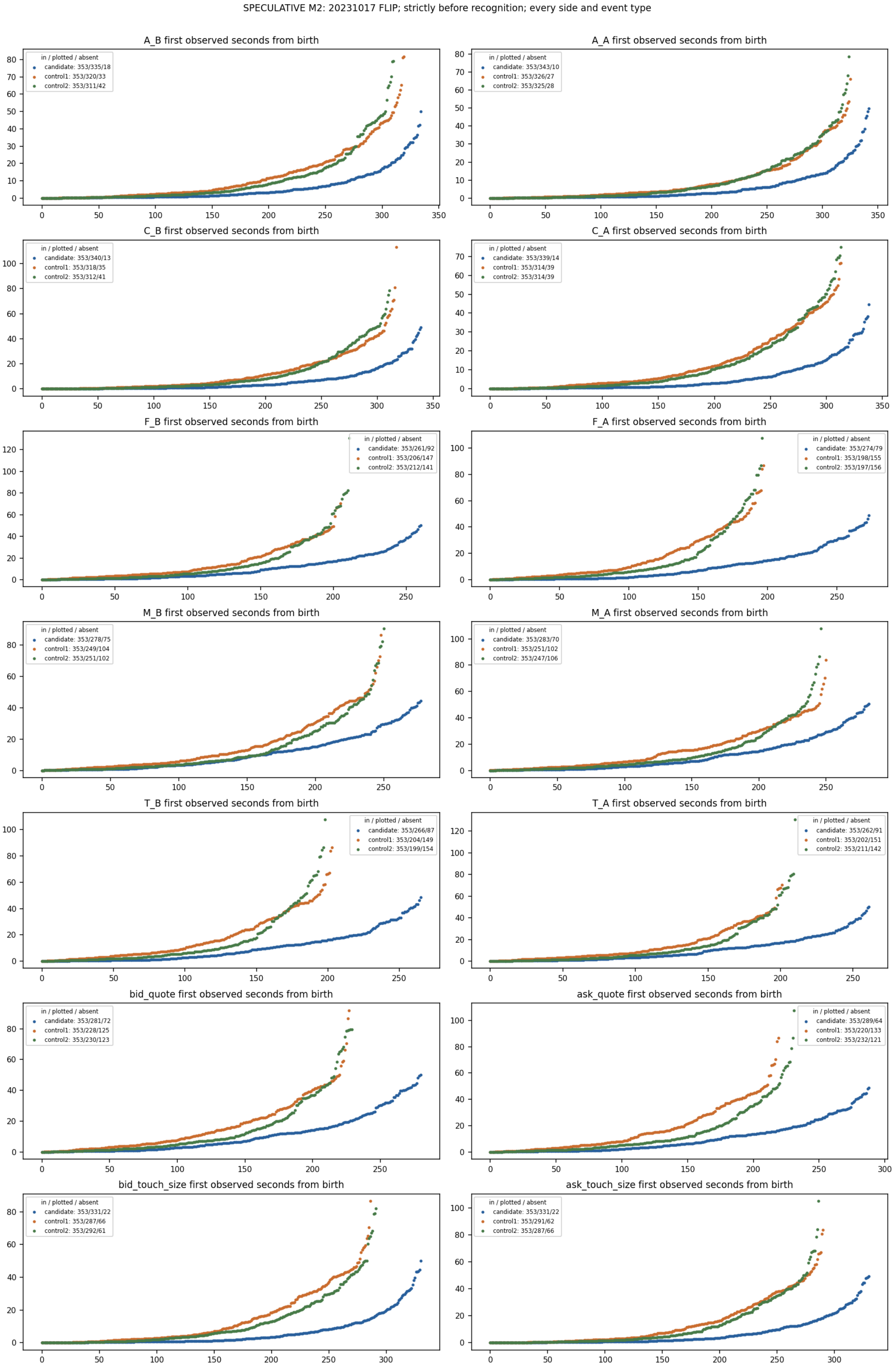
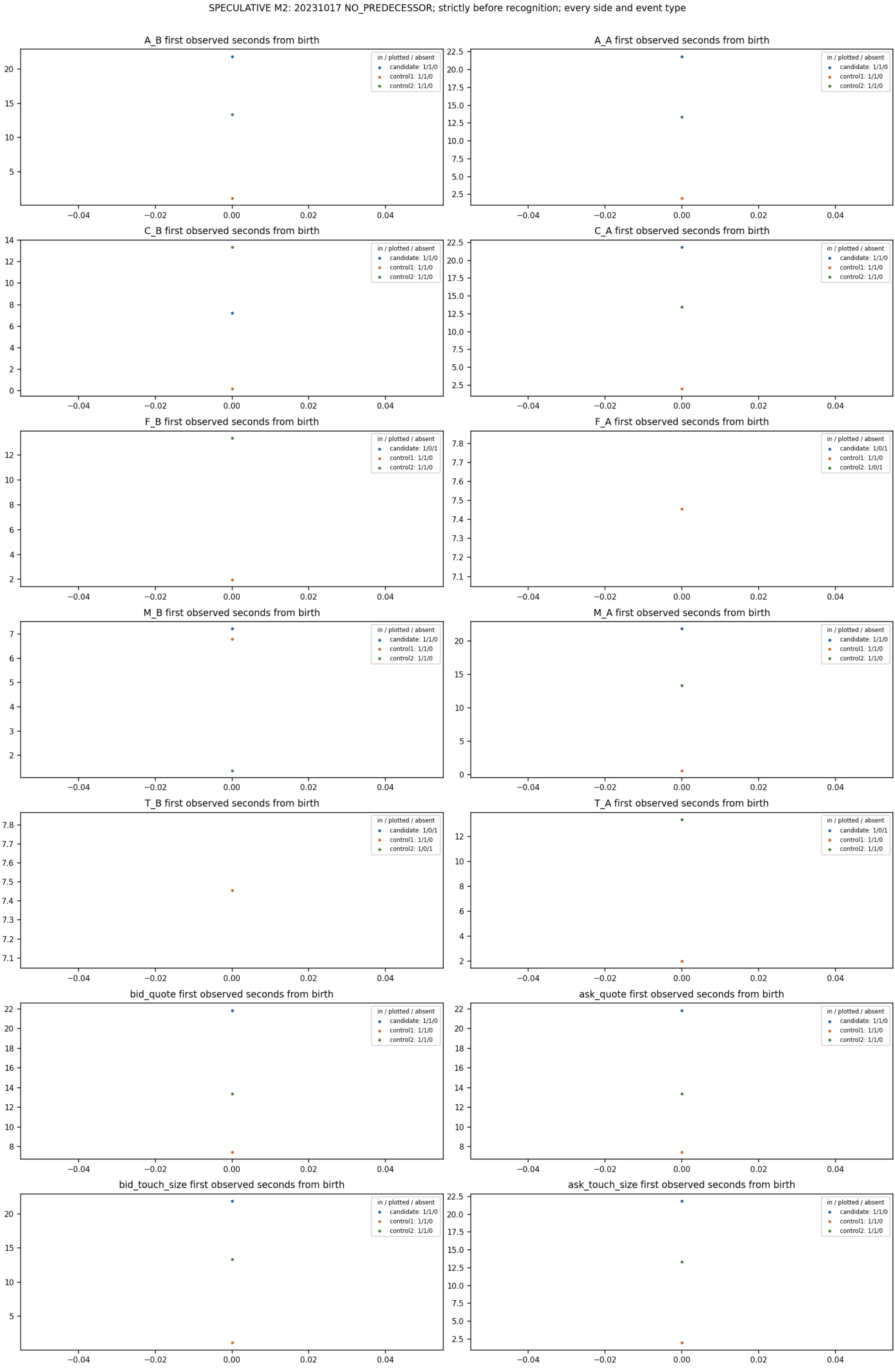
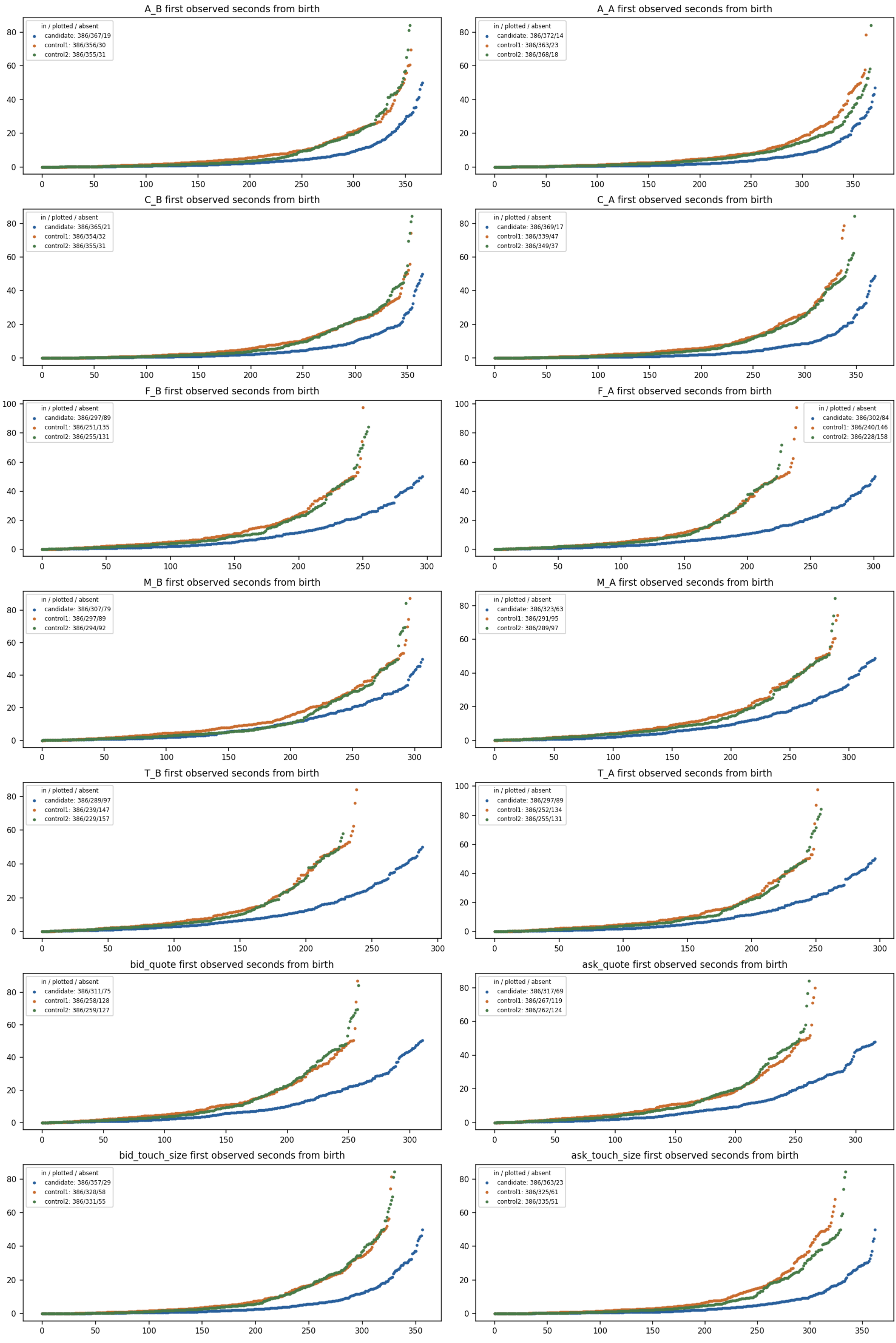
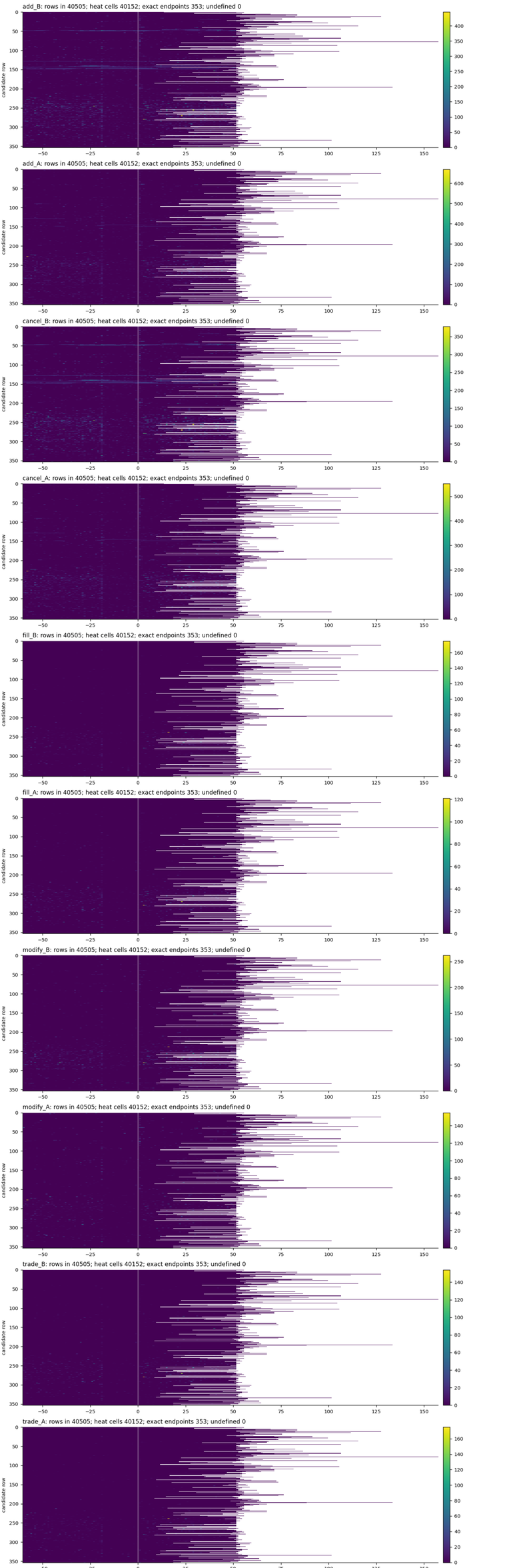

SPECULATIVE M2: 20231017 FLIP; strictly before recognition; every side and event type
A_B first observed seconds from birth
in / plotted / absent
candidate: 353/335/18
control1: 353/320/33
control2: 353/311/42
A_A first observed seconds from birth
in / plotted / absent
candidate: 353/343/10
control1: 353/326/27
control2: 353/325/28
C_B first observed seconds from birth
in / plotted / absent
candidate: 353/340/13
control1: 353/318/35
control2: 353/312/41
C_A first observed seconds from birth
in / plotted / absent
candidate: 353/339/14
control1: 353/314/39
control2: 353/314/39
F_B first observed seconds from birth
in / plotted / absent
candidate: 353/261/92
control1: 353/206/147
control2: 353/212/141
F_A first observed seconds from birth
in / plotted / absent
candidate: 353/274/79
control1: 353/198/155
control2: 353/197/156
M_B first observed seconds from birth
in / plotted / absent
candidate: 353/278/75
control1: 353/249/104
control2: 353/251/102
M_A first observed seconds from birth
in / plotted / absent
candidate: 353/283/70
control1: 353/251/102
control2: 353/247/106
T_B first observed seconds from birth
in / plotted / absent
candidate: 353/266/87
control1: 353/204/149
control2: 353/199/154
T_A first observed seconds from birth
in / plotted / absent
candidate: 353/262/91
control1: 353/202/151
control2: 353/211/142
bid_quote first observed seconds from birth
in / plotted / absent
candidate: 353/281/72
control1: 353/228/125
control2: 353/230/123
ask_quote first observed seconds from birth
in / plotted / absent
candidate: 353/289/64
control1: 353/220/133
control2: 353/232/121
bid_touch_size first observed seconds from birth
in / plotted / absent
candidate: 353/331/22
control1: 353/287/66
control2: 353/292/61
ask_touch_size first observed seconds from birth
in / plotted / absent
candidate: 353/331/22
control1: 353/291/62
control2: 353/287/66


SPECULATIVE M2: 20231017 NO_PREDECESSOR; strictly before recognition; every side and event type
A_B first observed seconds from birth
A_A first observed seconds from birth
C_B first observed seconds from birth
C_A first observed seconds from birth
F_B first observed seconds from birth
F_A first observed seconds from birth
M_B first observed seconds from birth
M_A first observed seconds from birth
T_B first observed seconds from birth
T_A first observed seconds from birth
bid_quote first observed seconds from birth
ask_quote first observed seconds from birth
bid_touch_size first observed seconds from birth
ask_touch_size first observed seconds from birth
in / plotted / absent
candidate: 1/1/0
control1: 1/1/0
control2: 1/1/0
candidate: 1/0/1
control2: 1/0/1
candidate: 1/0/1
control2: 1/0/1

SPECULATIVE M2: 20231017 SAME; strictly before recognition; every side and event type

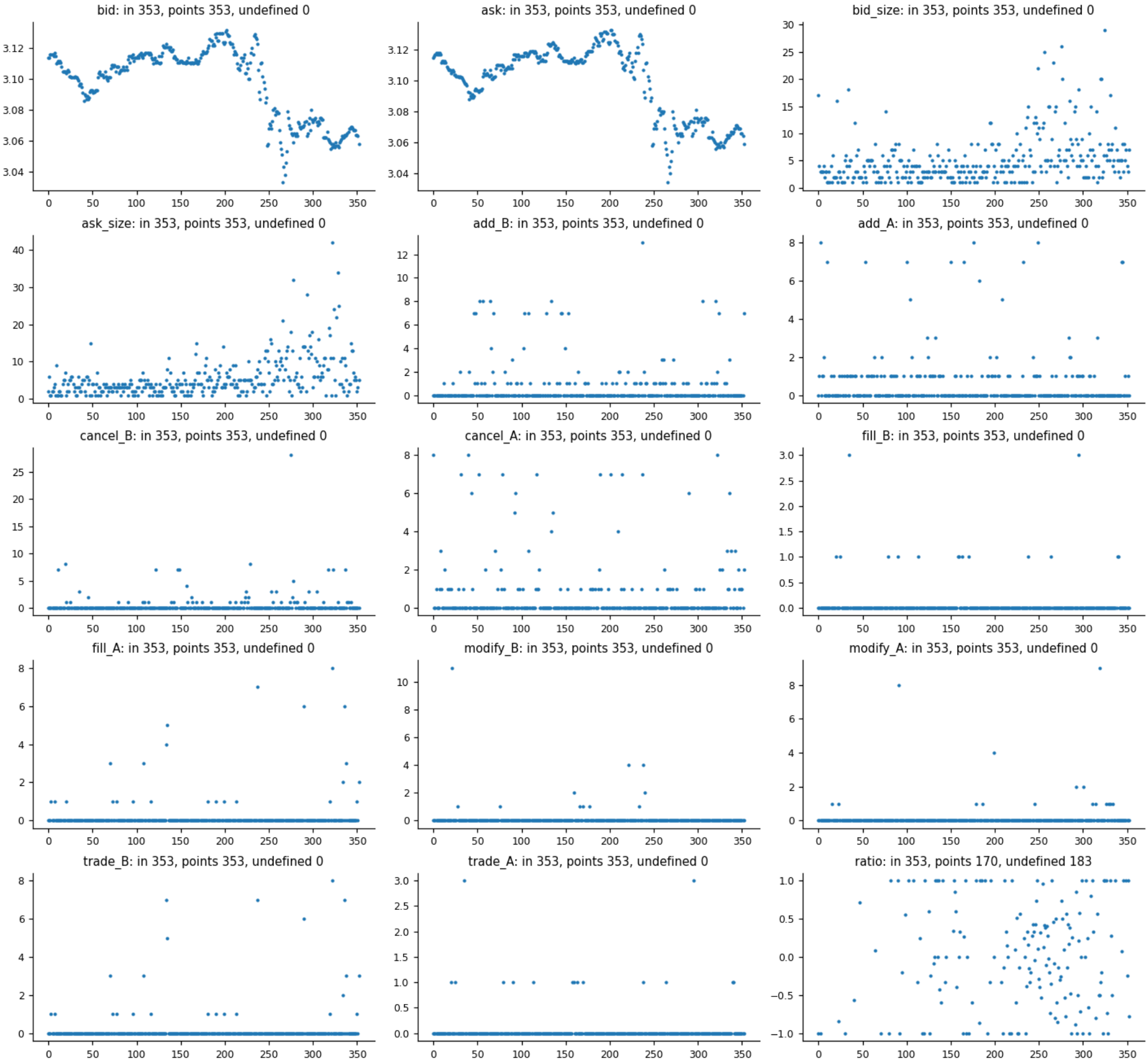


20231017 candidate FLIP: raw levels and lots; 852 subset on Oct 18

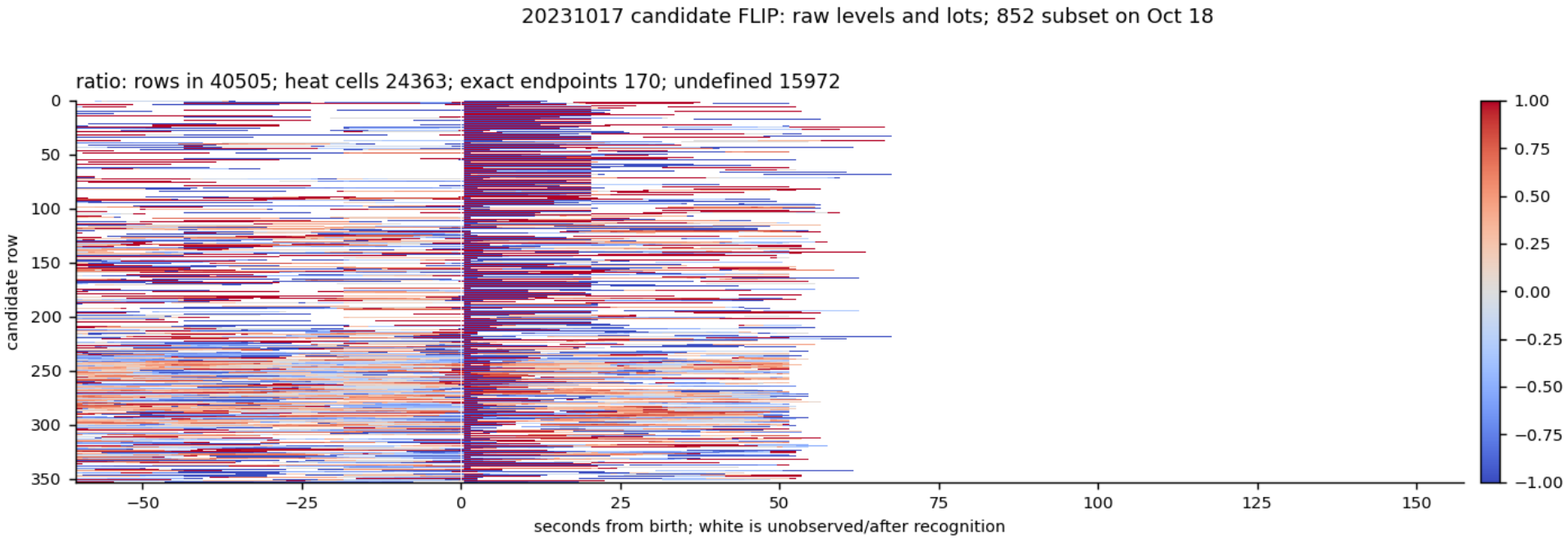


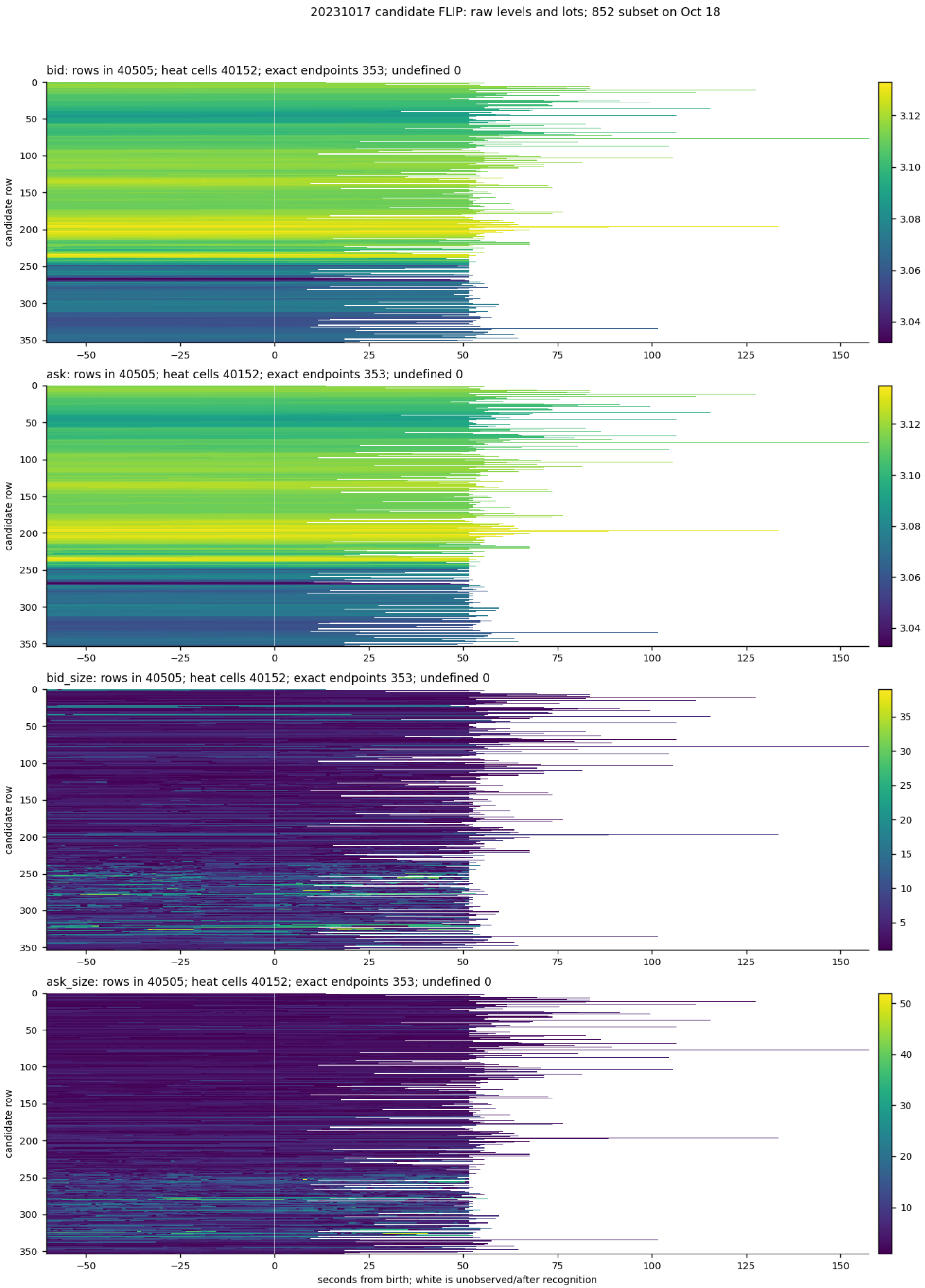
20231017-candidate-FLIP-endpoints.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



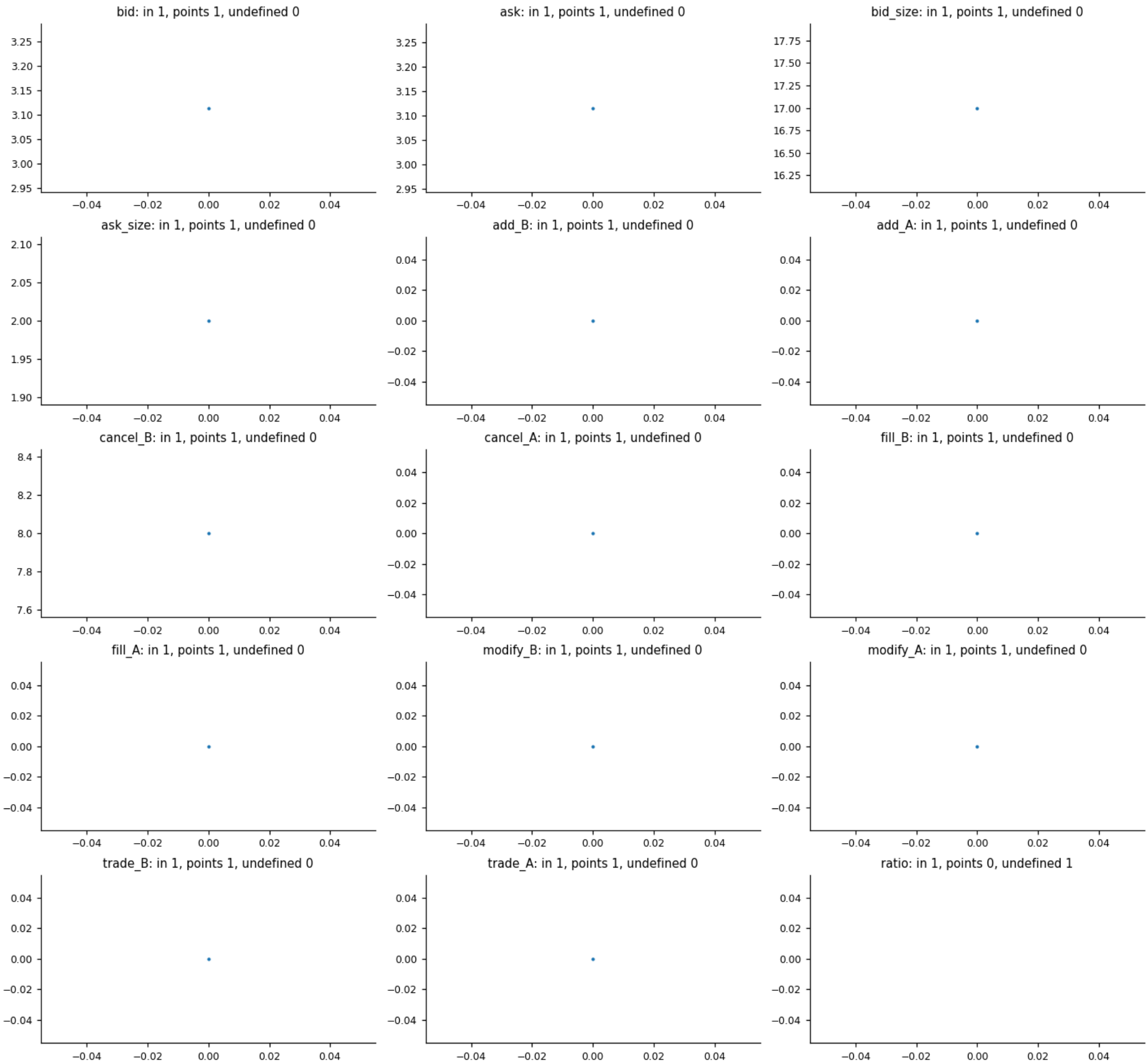

20231017 candidate FLIP: raw levels and lots; 852 subset on Oct 18
bid: rows in 40505; heat cells 40152; exact endpoints 353; undefined 0
ask: rows in 40505; heat cells 40152; exact endpoints 353; undefined 0
bid_size: rows in 40505; heat cells 40152; exact endpoints 353; undefined 0
ask_size: rows in 40505; heat cells 40152; exact endpoints 353; undefined 0
candidate row
seconds from birth; white is unobserved/after recognition

20231017-candidate-NO_PREDECESSOR-actions.png

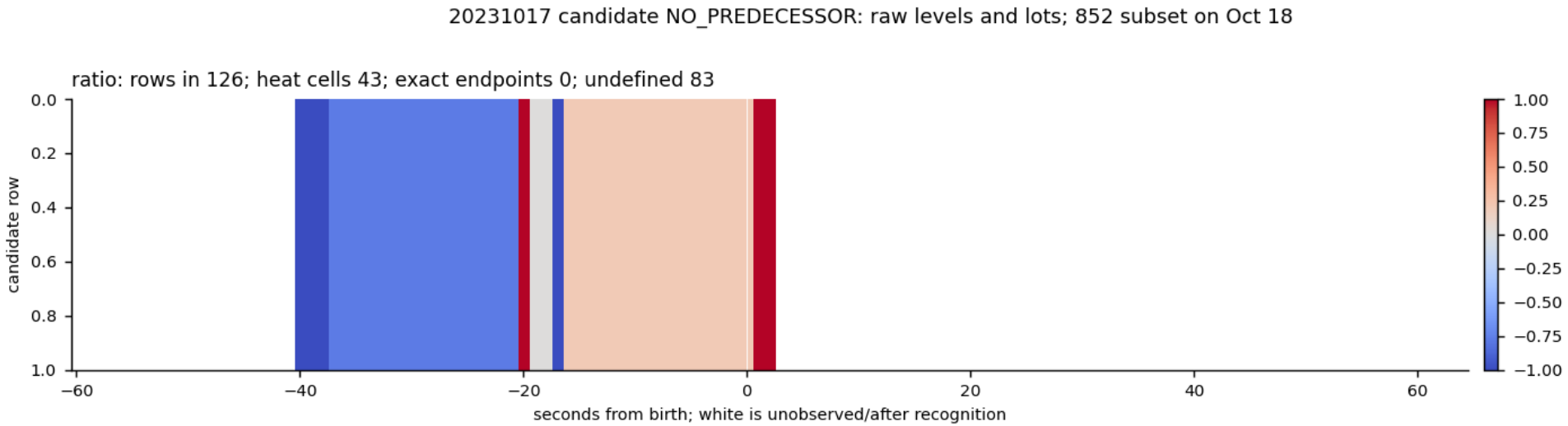
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

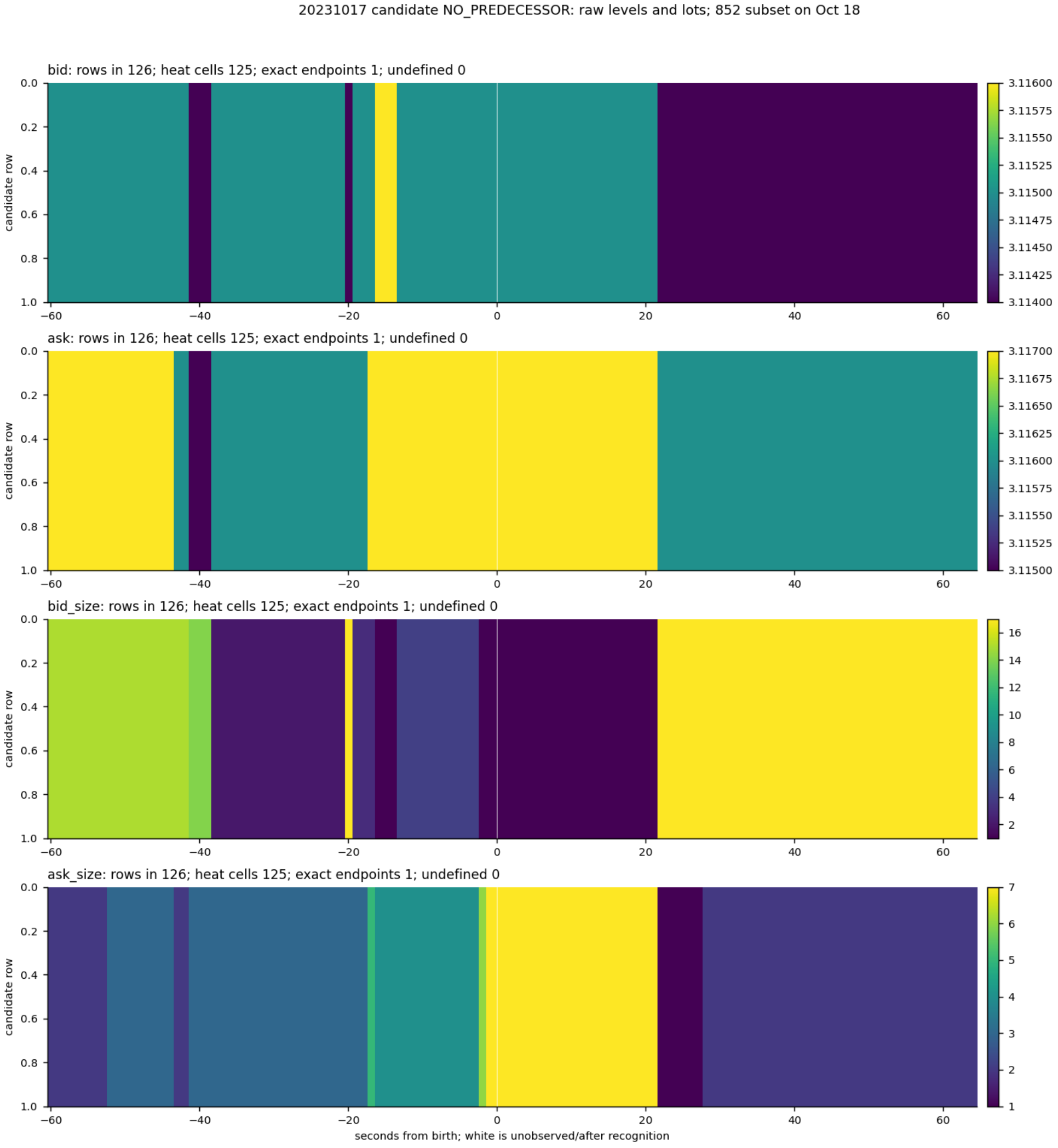
20231017-candidate-NO_PREDECESSOR-endpoints.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

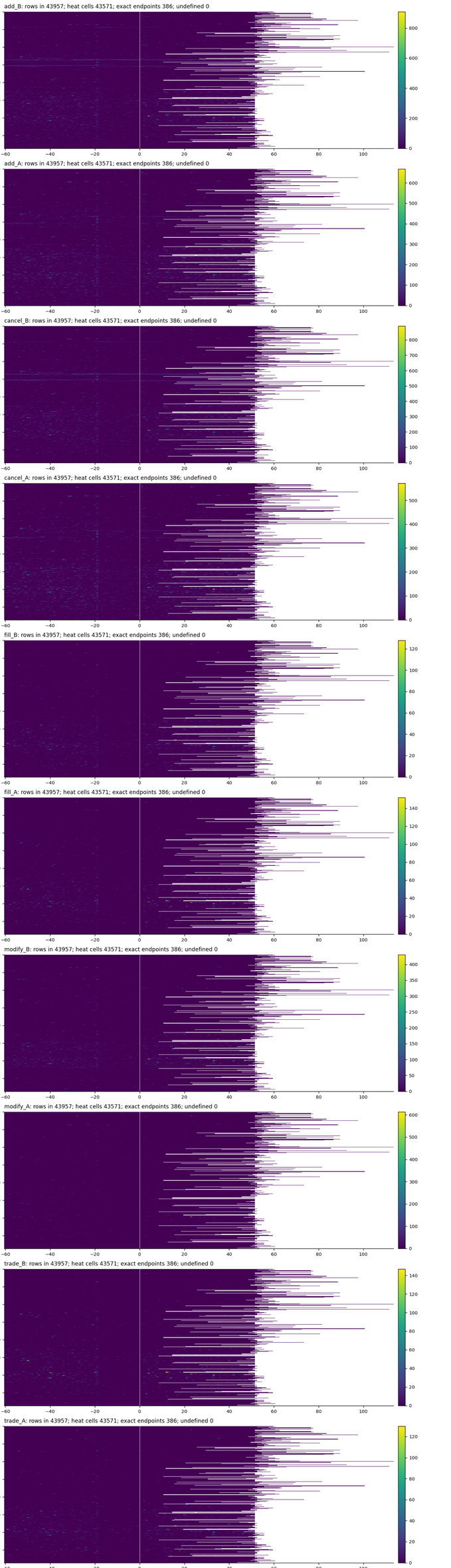


Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



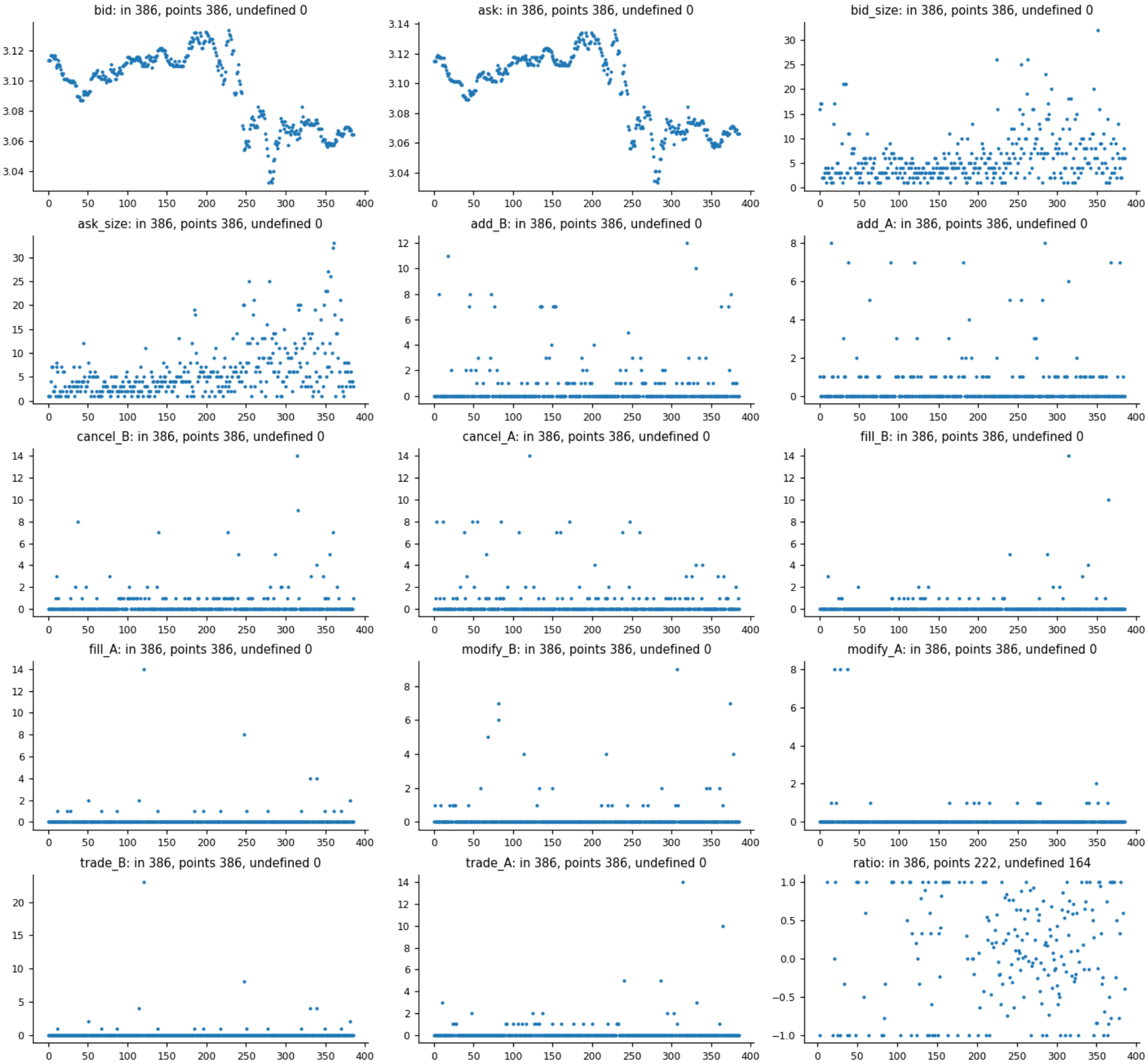

20231017 candidate NO_PREDECESSOR: raw levels and lots; 852 subset on Oct 18
bid: rows in 126; heat cells 125; exact endpoints 1; undefined 0
ask: rows in 126; heat cells 125; exact endpoints 1; undefined 0
bid_size: rows in 126; heat cells 125; exact endpoints 1; undefined 0
ask_size: rows in 126; heat cells 125; exact endpoints 1; undefined 0
candidate row
seconds from birth; white is unobserved/after recognition

20231017 candidate SAME: raw levels and lots; 852 subset on Oct 18

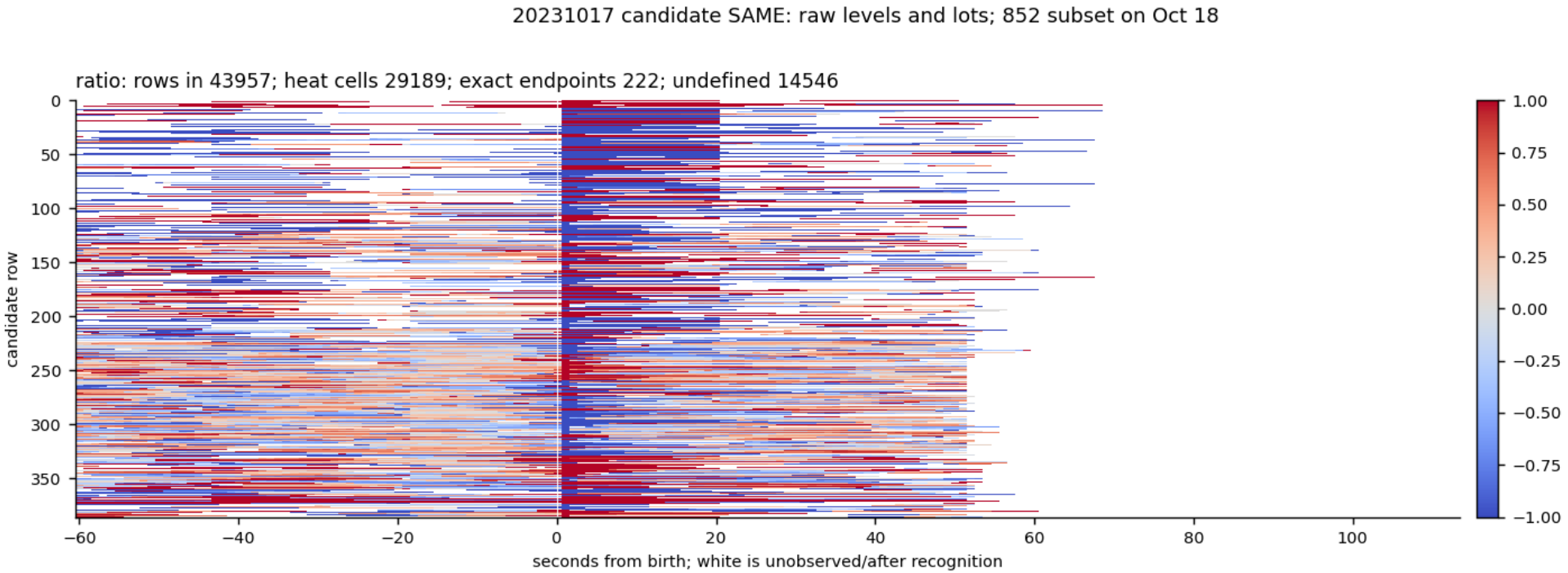


20231017-candidate-SAME-endpoints.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

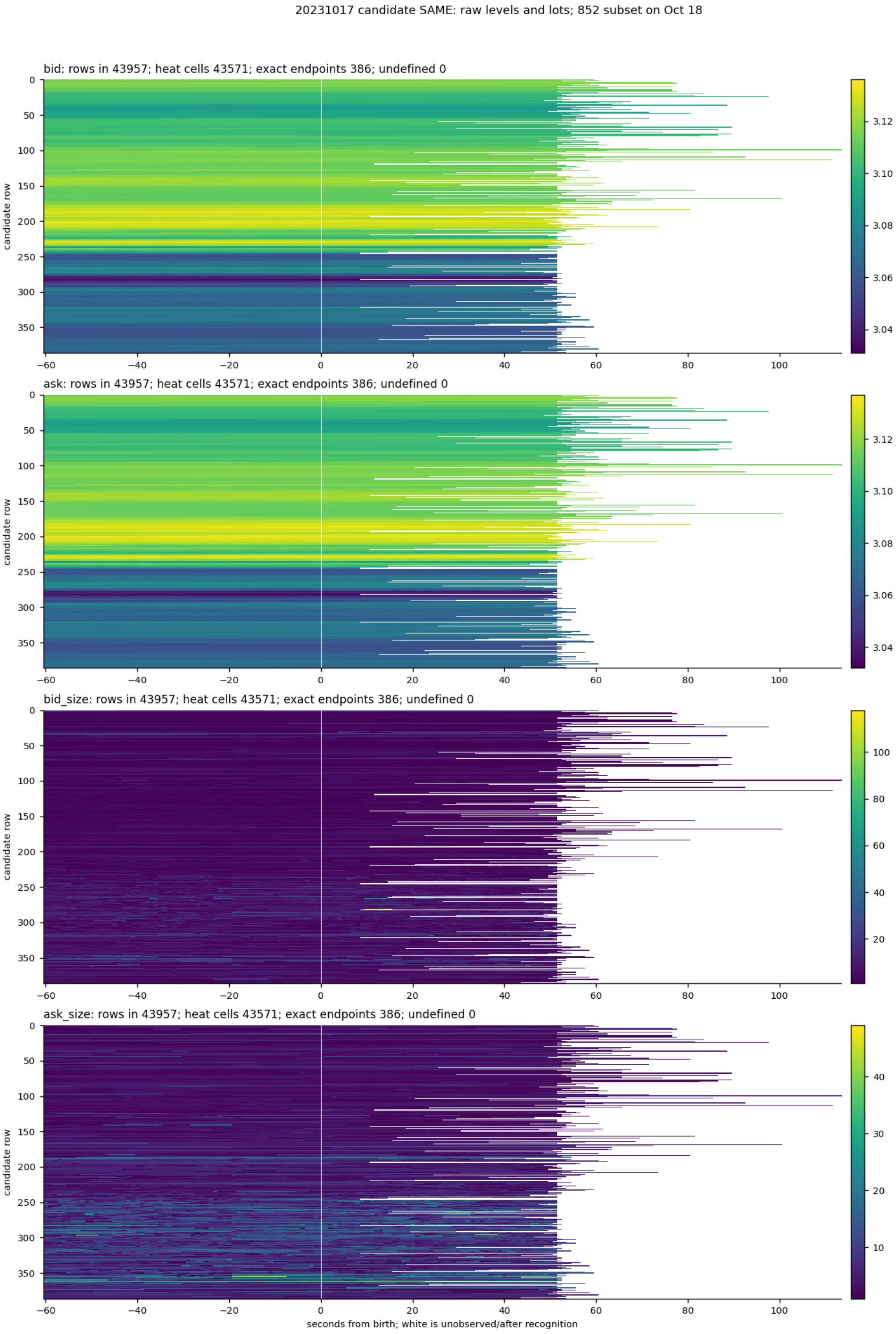


Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

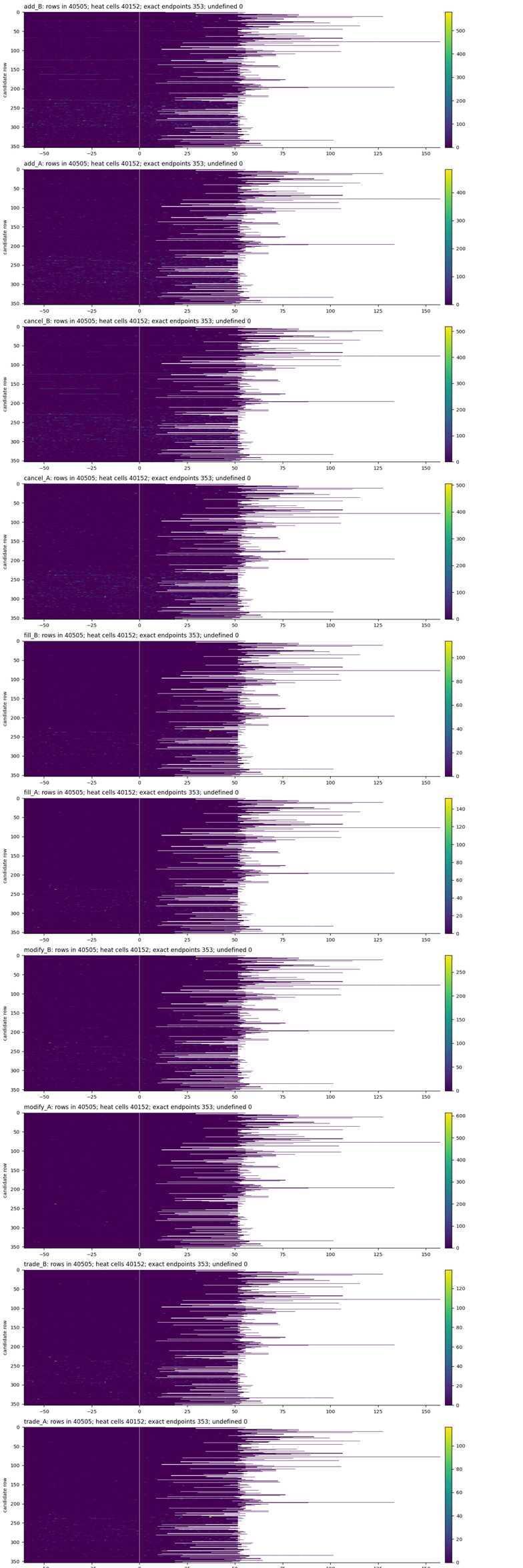


20231017-candidate-SAME-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

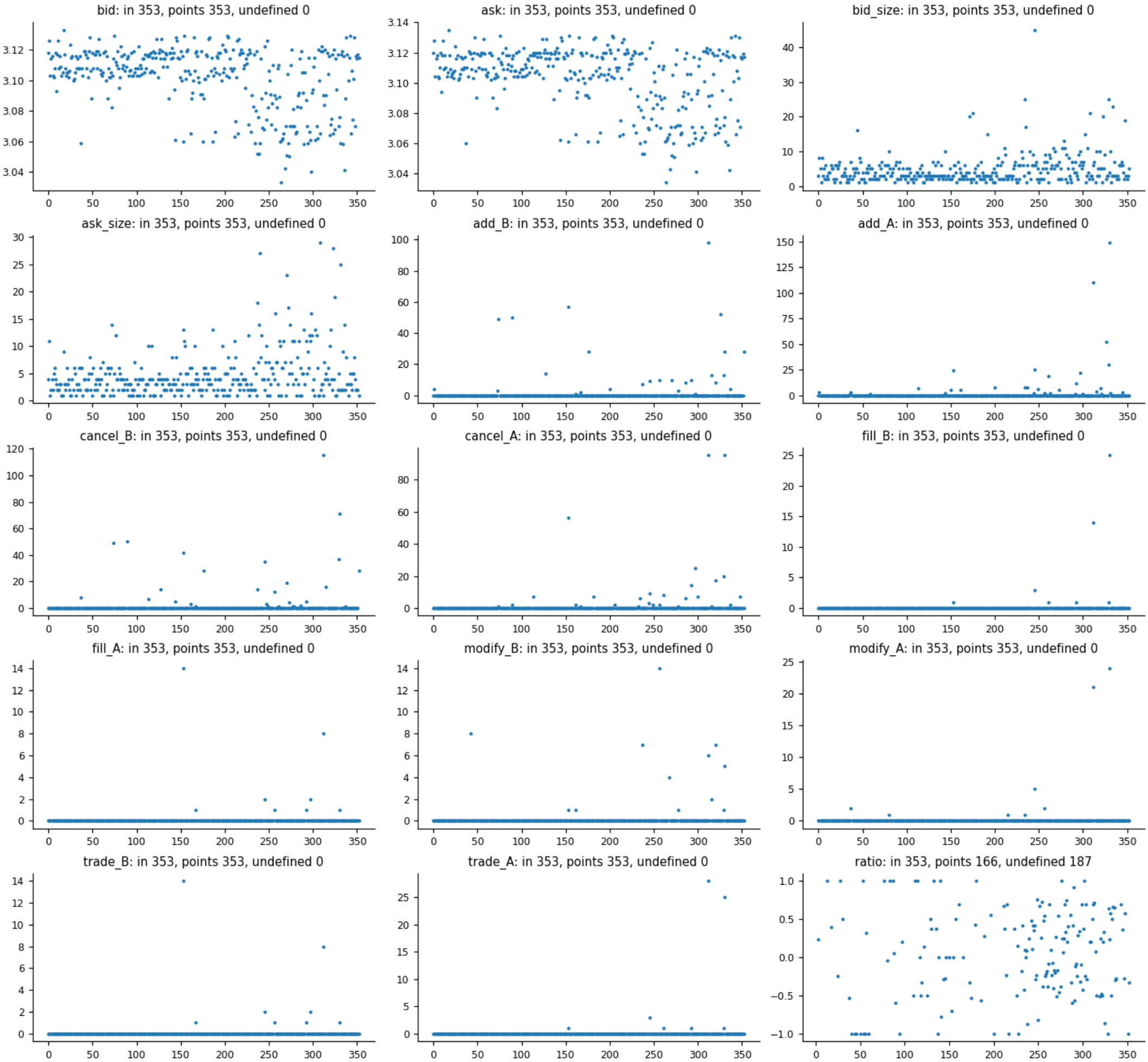


20231017 control1 FLIP: raw levels and lots; 852 subset on Oct 18

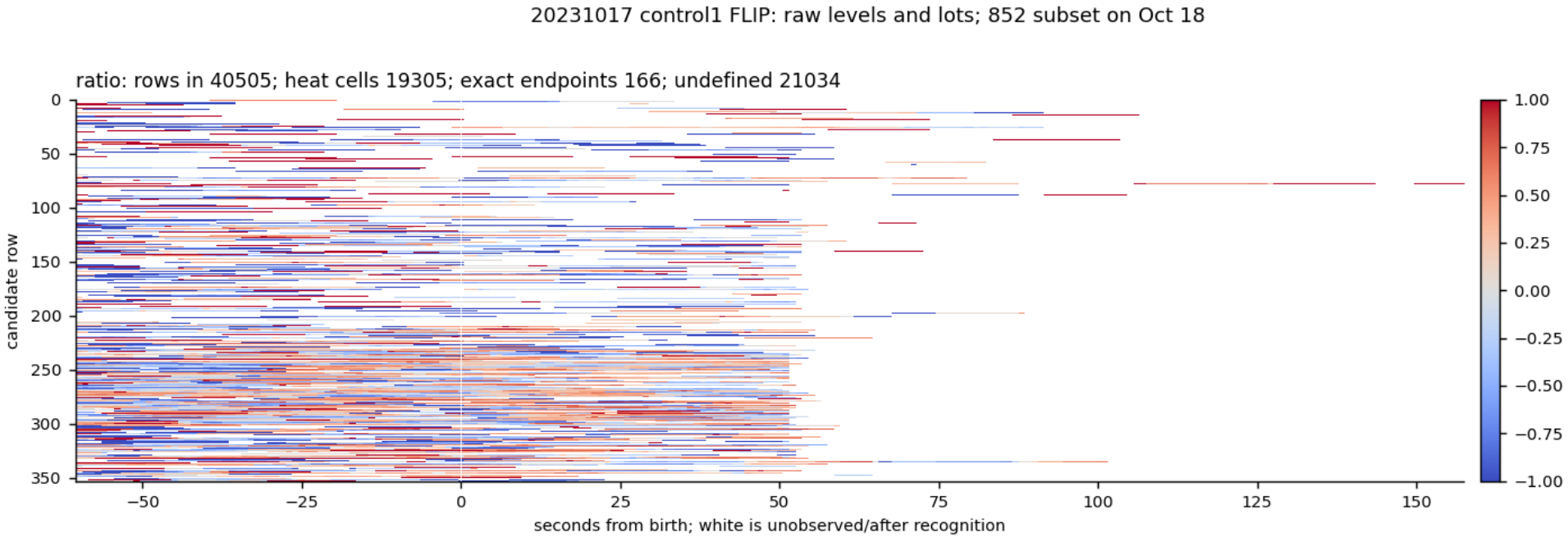


20231017-control1-FLIP-endpoints.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

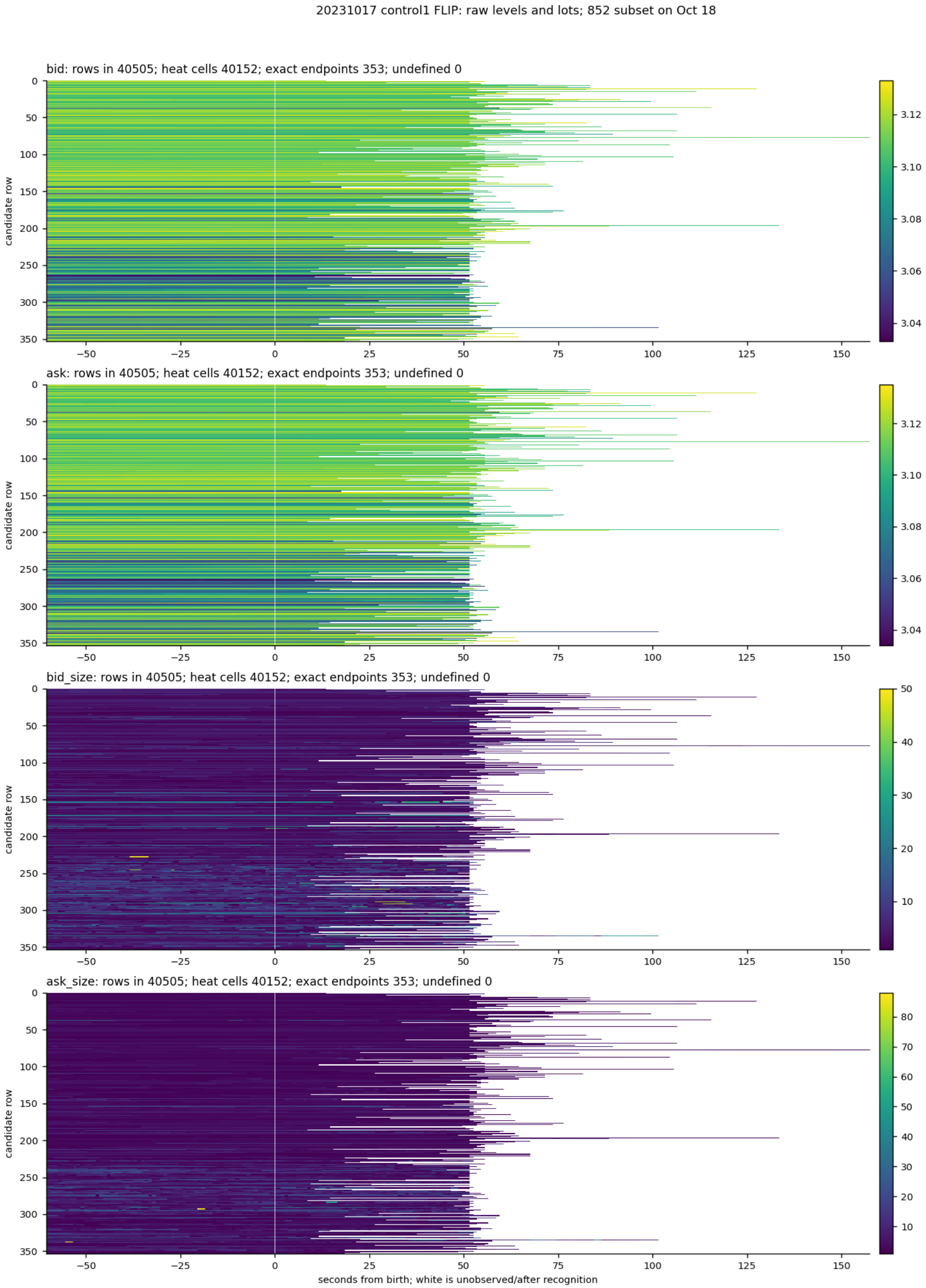


Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



20231017-control1-FLIP-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

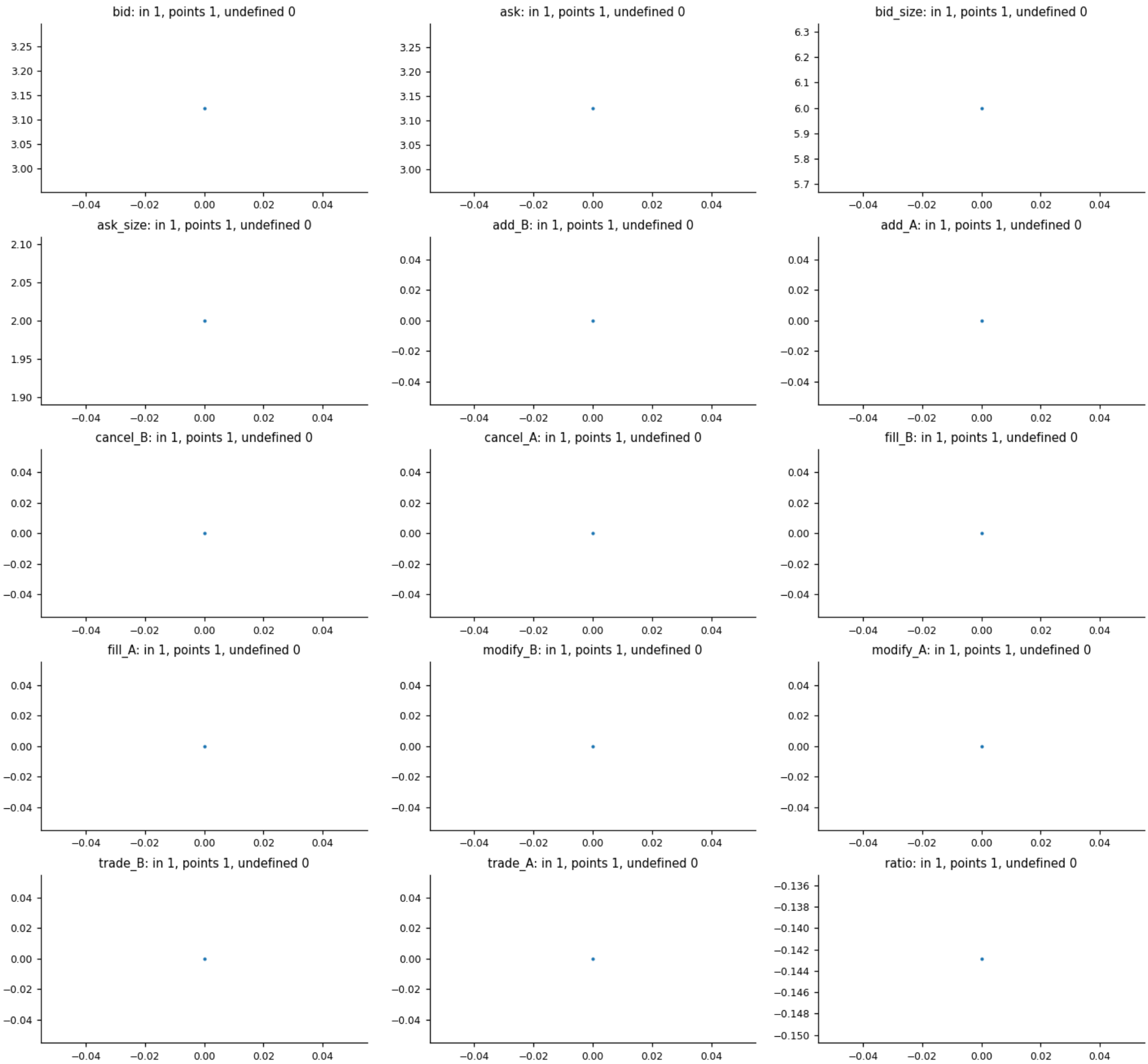


20231017-control1-NO_PREDECESSOR-actions.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

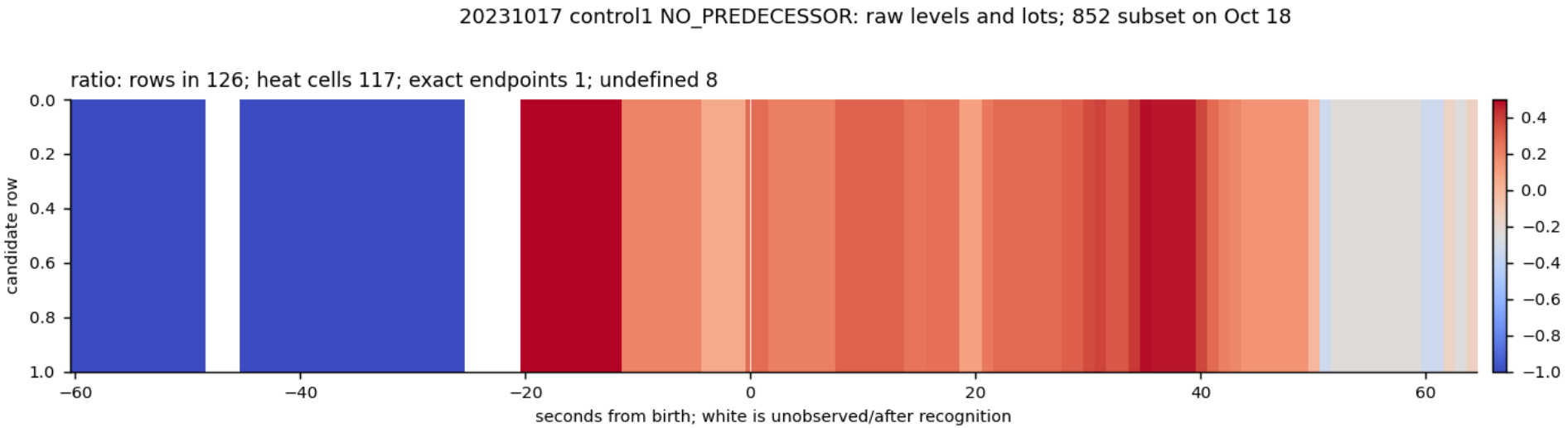
20231017-control1-NO_PREDECESSOR-endpoints.png

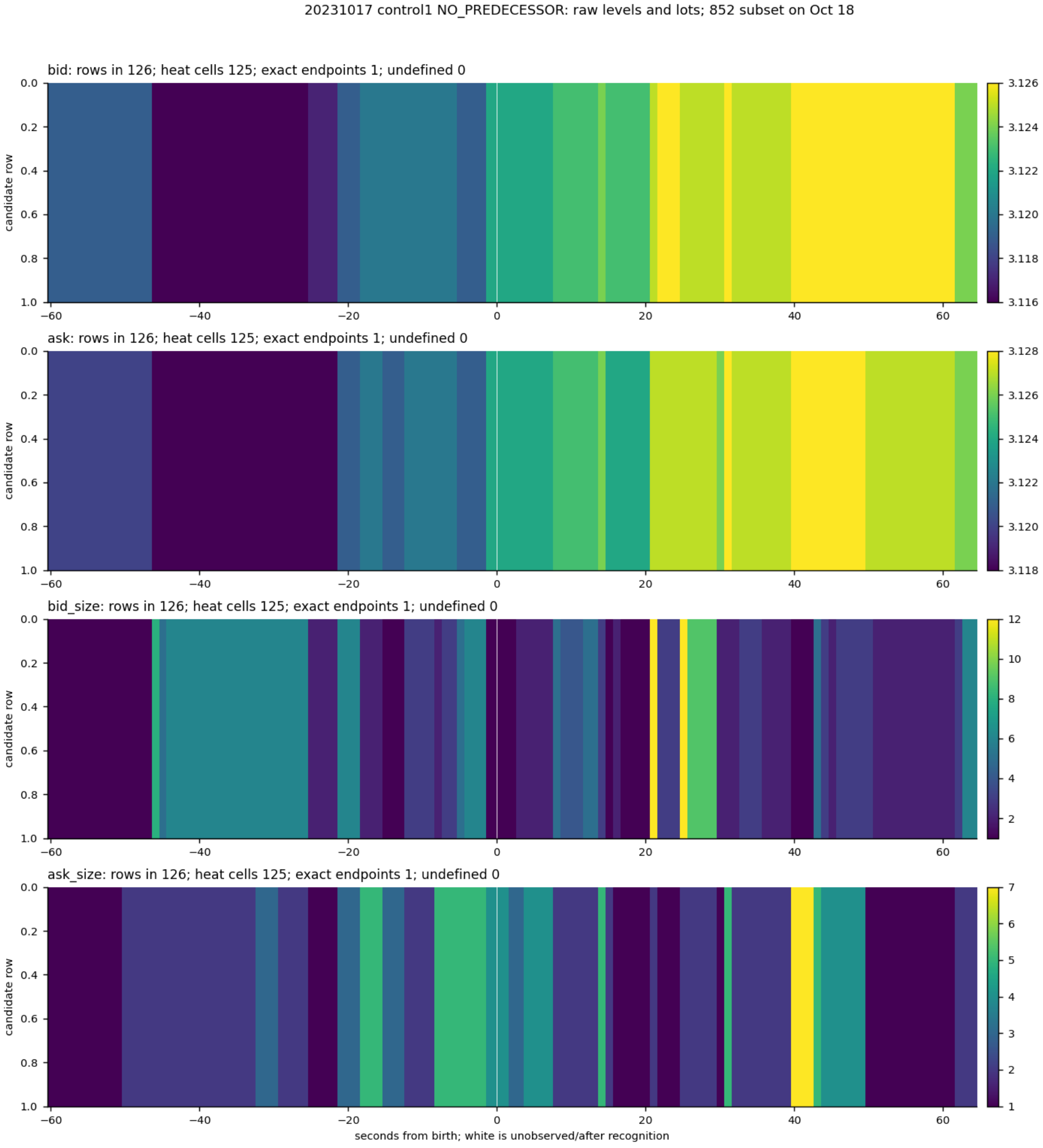
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



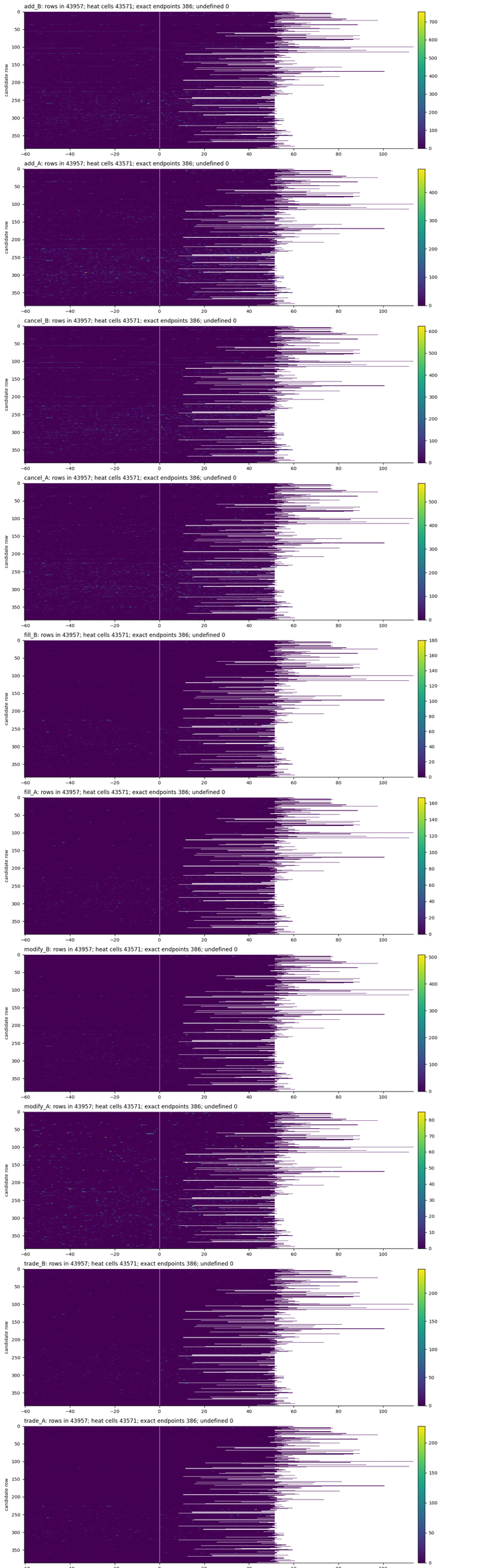
20231017-control1-NO_PREDECESSOR-flow.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



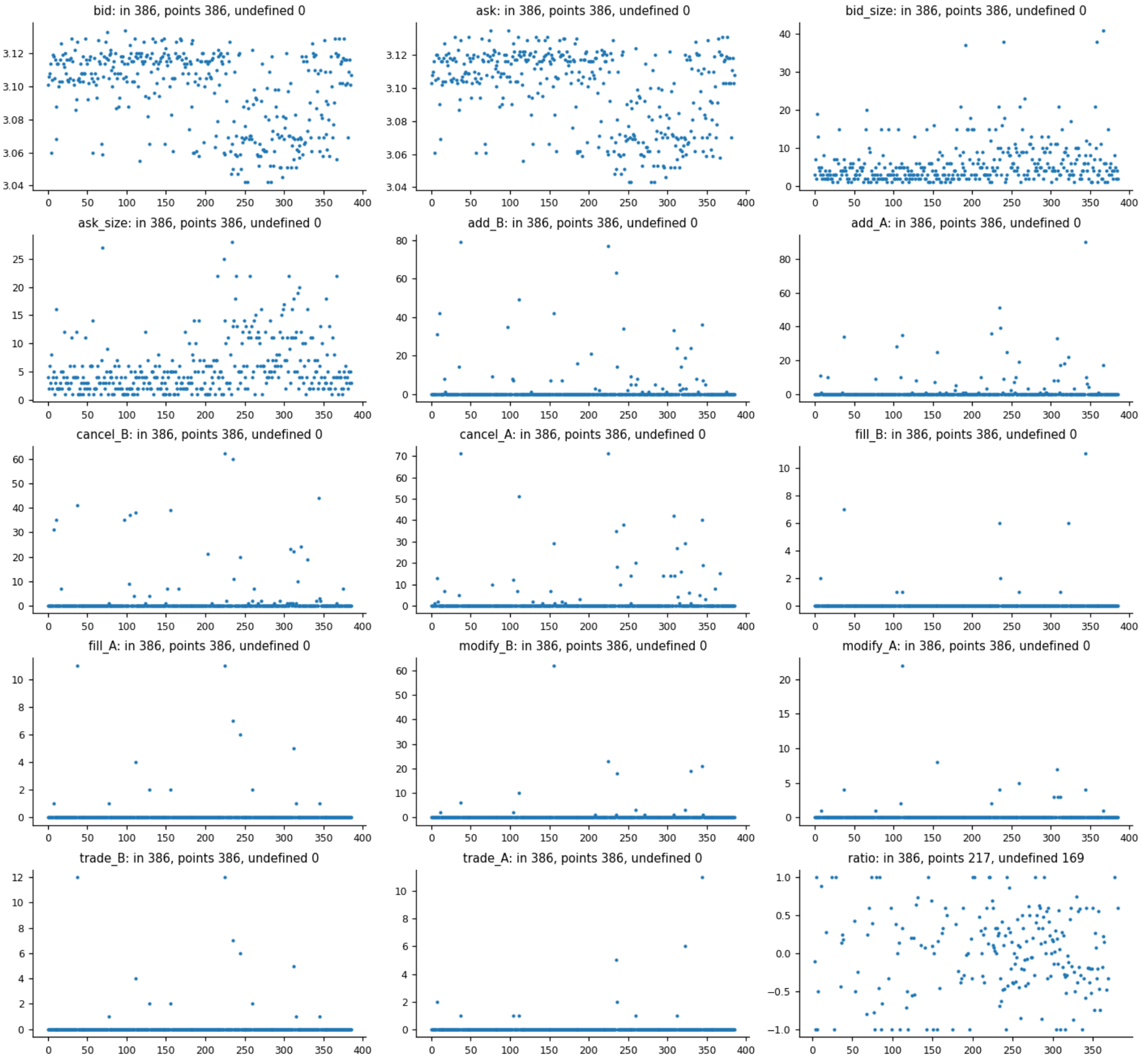

20231017 control1 NO_PREDECESSOR: raw levels and lots; 852 subset on Oct 18
bid: rows in 126; heat cells 125; exact endpoints 1; undefined 0
ask: rows in 126; heat cells 125; exact endpoints 1; undefined 0
bid_size: rows in 126; heat cells 125; exact endpoints 1; undefined 0
ask_size: rows in 126; heat cells 125; exact endpoints 1; undefined 0
candidate row
seconds from birth; white is unobserved/after recognition

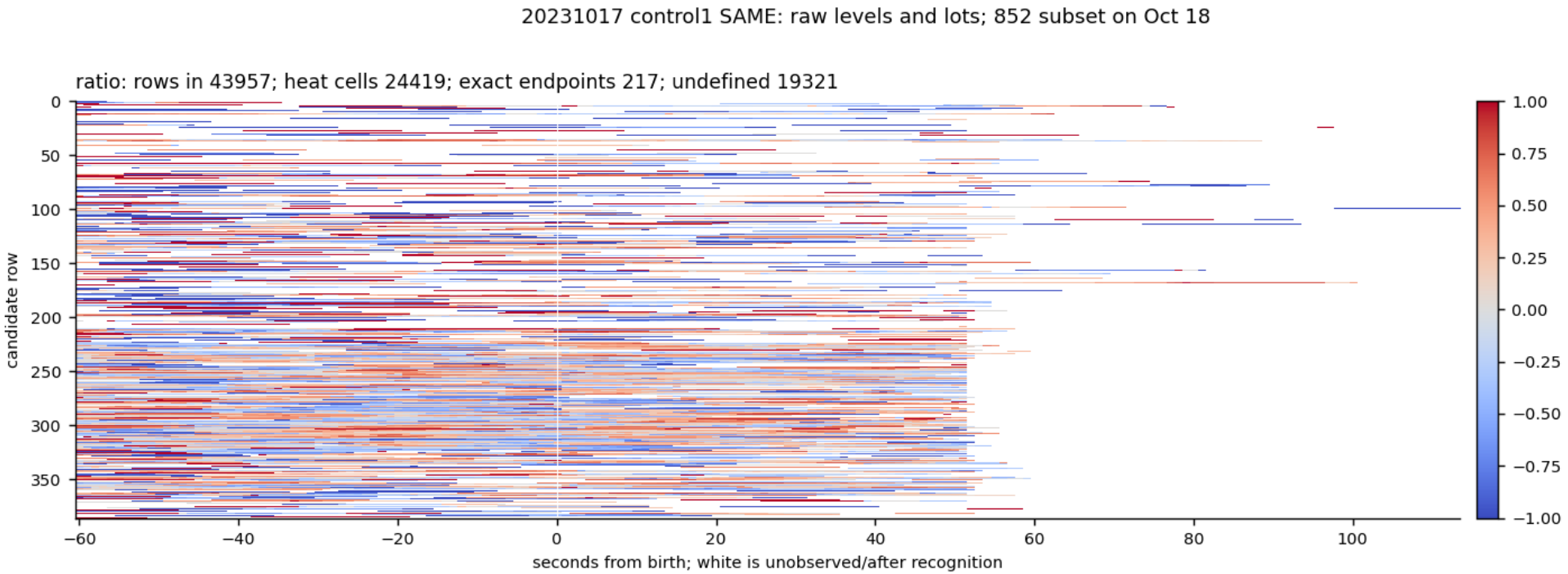
20231017 control1 SAME: raw levels and lots; 852 subset on Oct 18



20231017-control1-SAME-endpoints.png

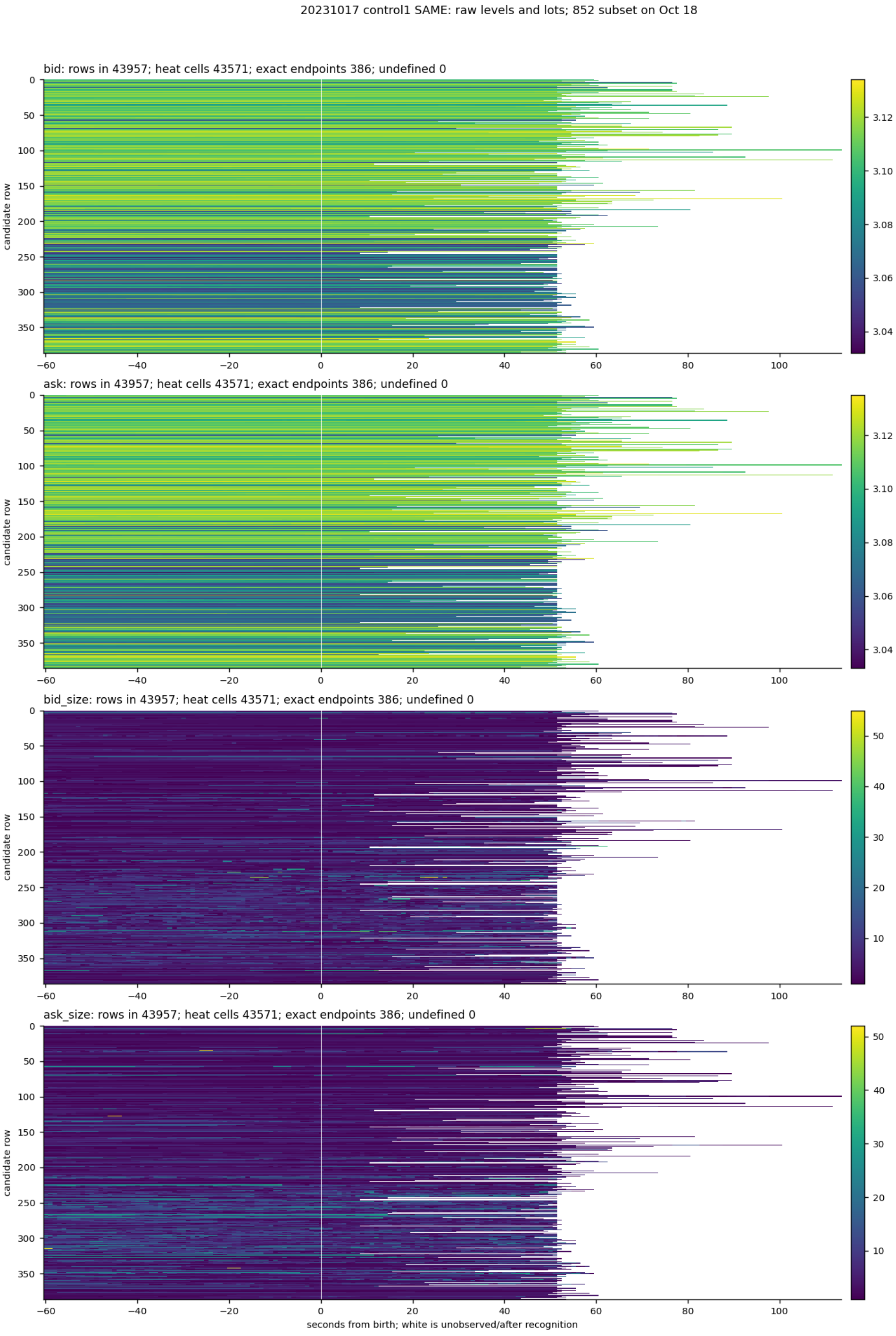
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



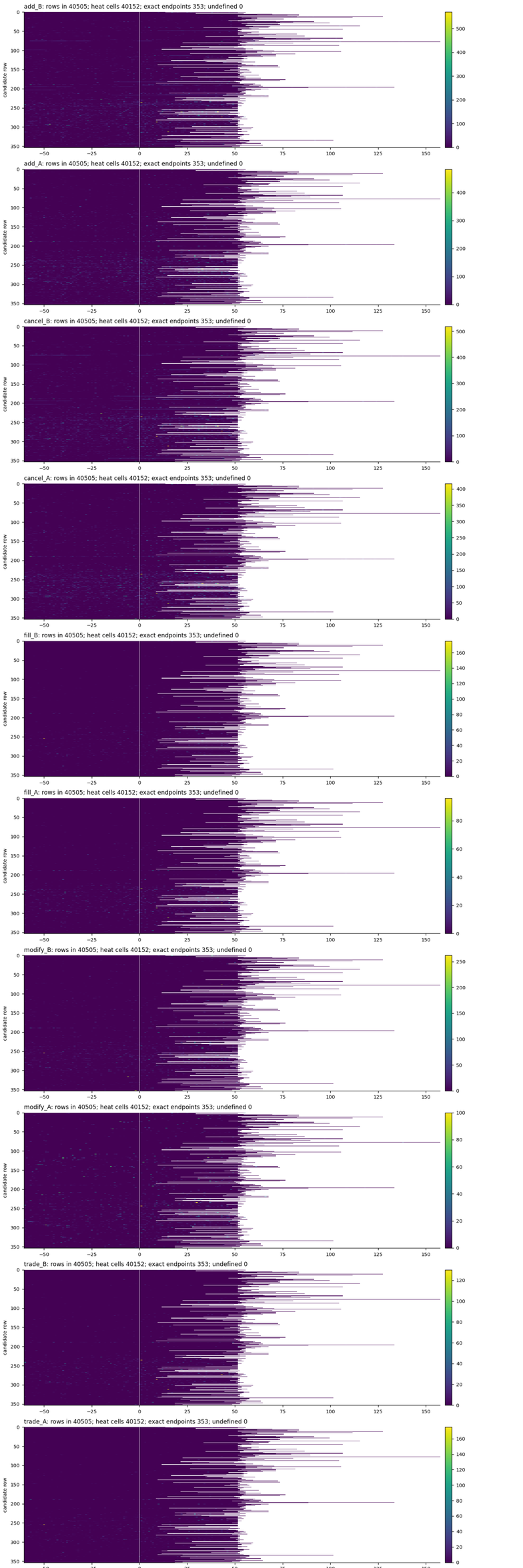

20231017 control1 SAME: raw levels and lots; 852 subset on Oct 18
ratio: rows in 43957; heat cells 24419; exact endpoints 217; undefined 19321
candidate row
0
50
100
150
200
250
300
350
−60
−40
−20
0
20
40
60
80
100
seconds from birth; white is unobserved/after recognition
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

20231017-control1-SAME-quotes.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

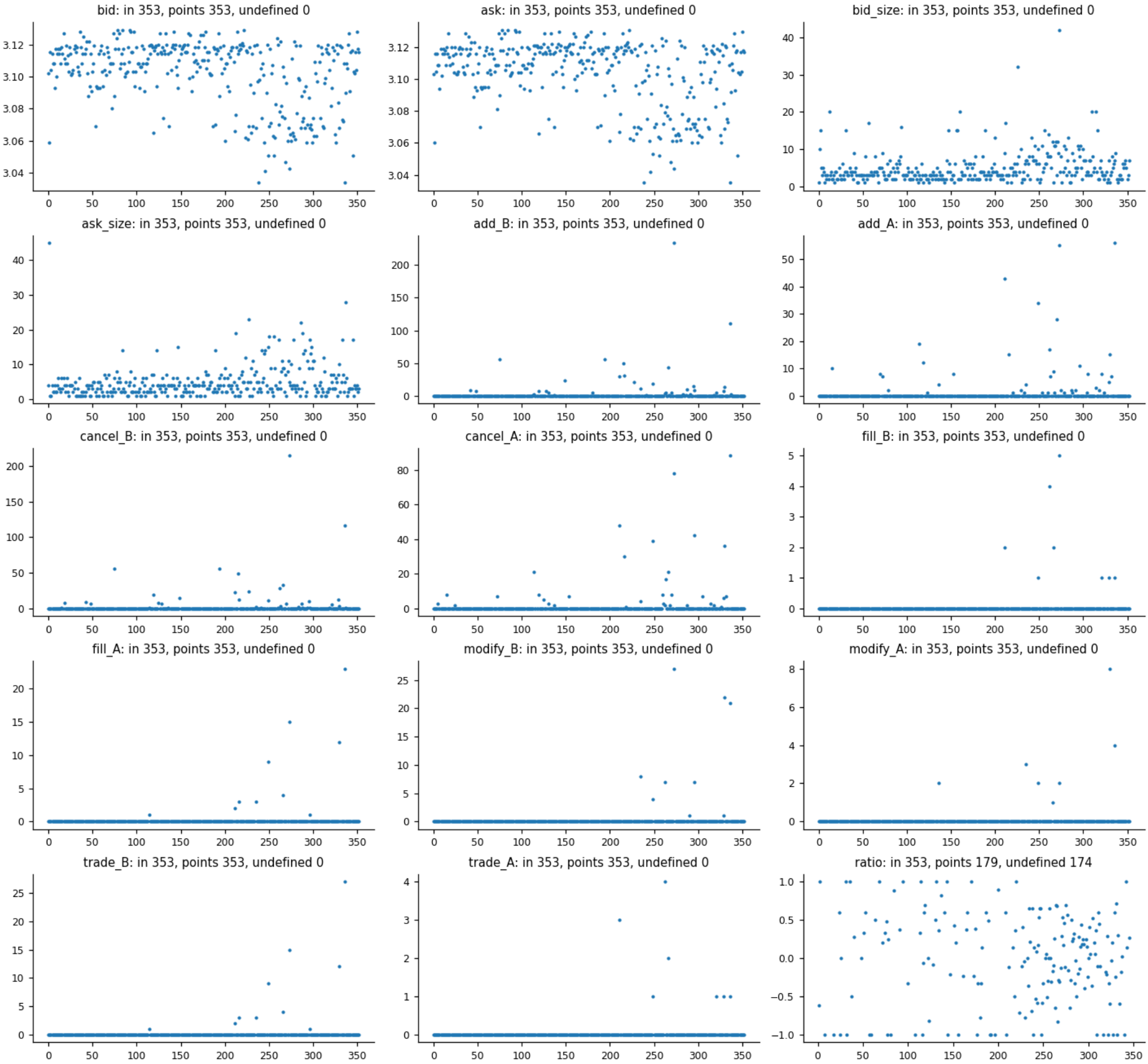


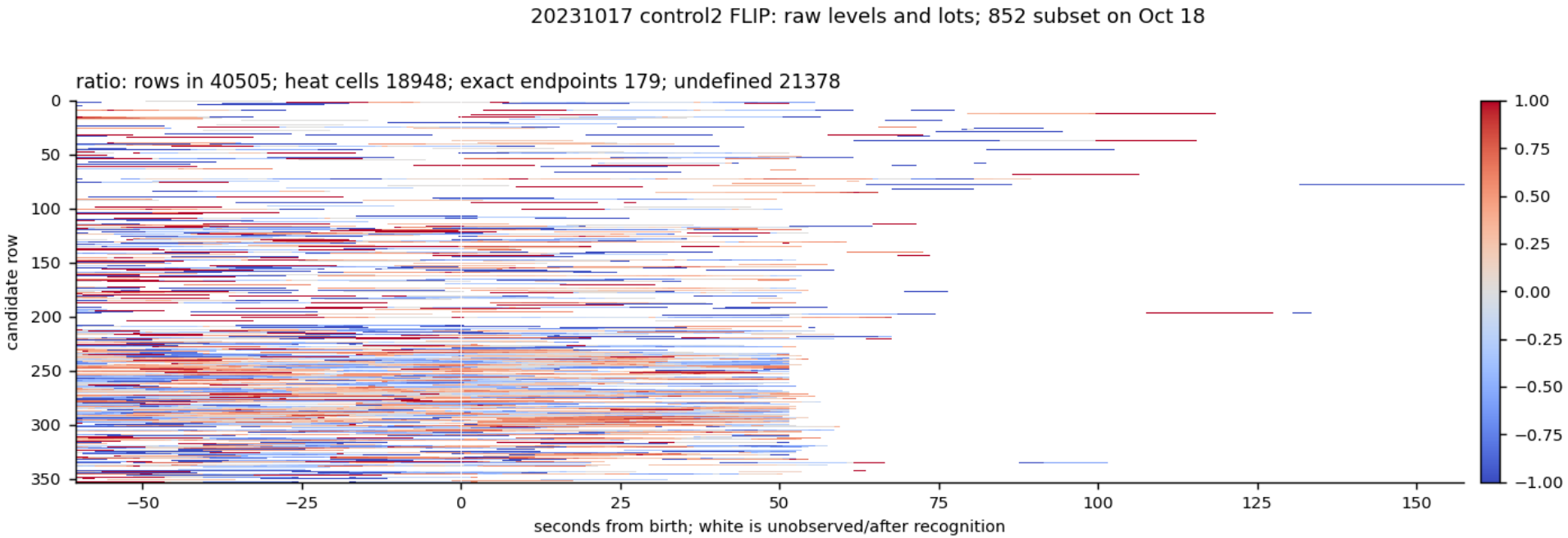
20231017 control2 FLIP: raw levels and lots; 852 subset on Oct 18



20231017-control2-FLIP-endpoints.png

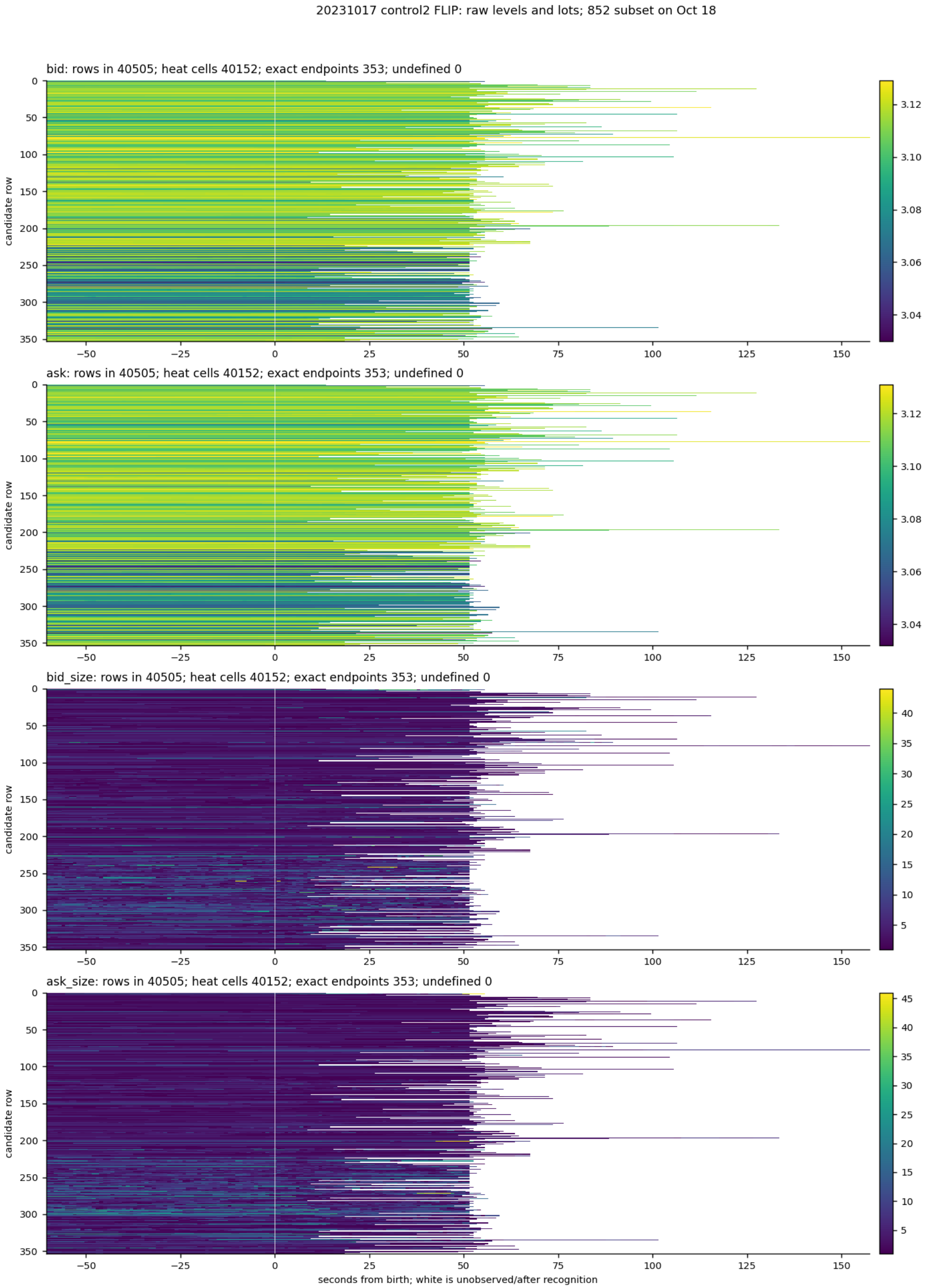
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

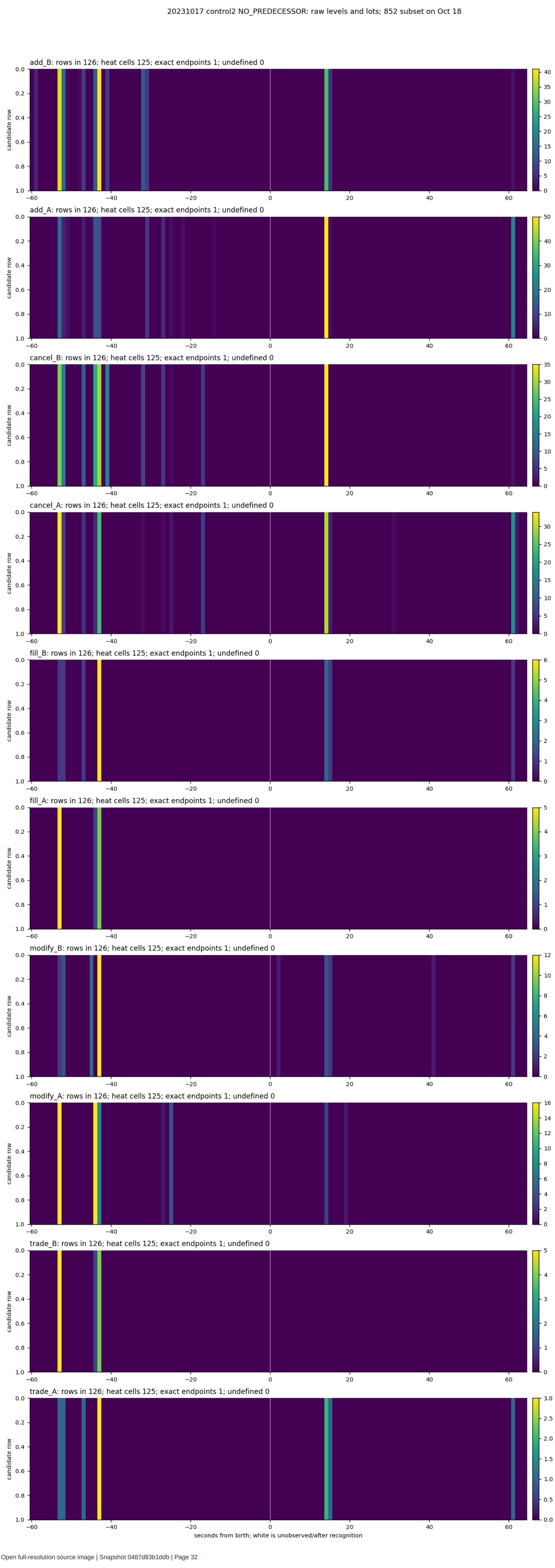



20231017 control2 FLIP: raw levels and lots; 852 subset on Oct 18
ratio: rows in 40505; heat cells 18948; exact endpoints 179; undefined 21378
candidate row
0
50
100
150
200
250
300
350
−50
−25
0
25
50
75
100
125
150
seconds from birth; white is unobserved/after recognition
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

20231017-control2-FLIP-quotes.png

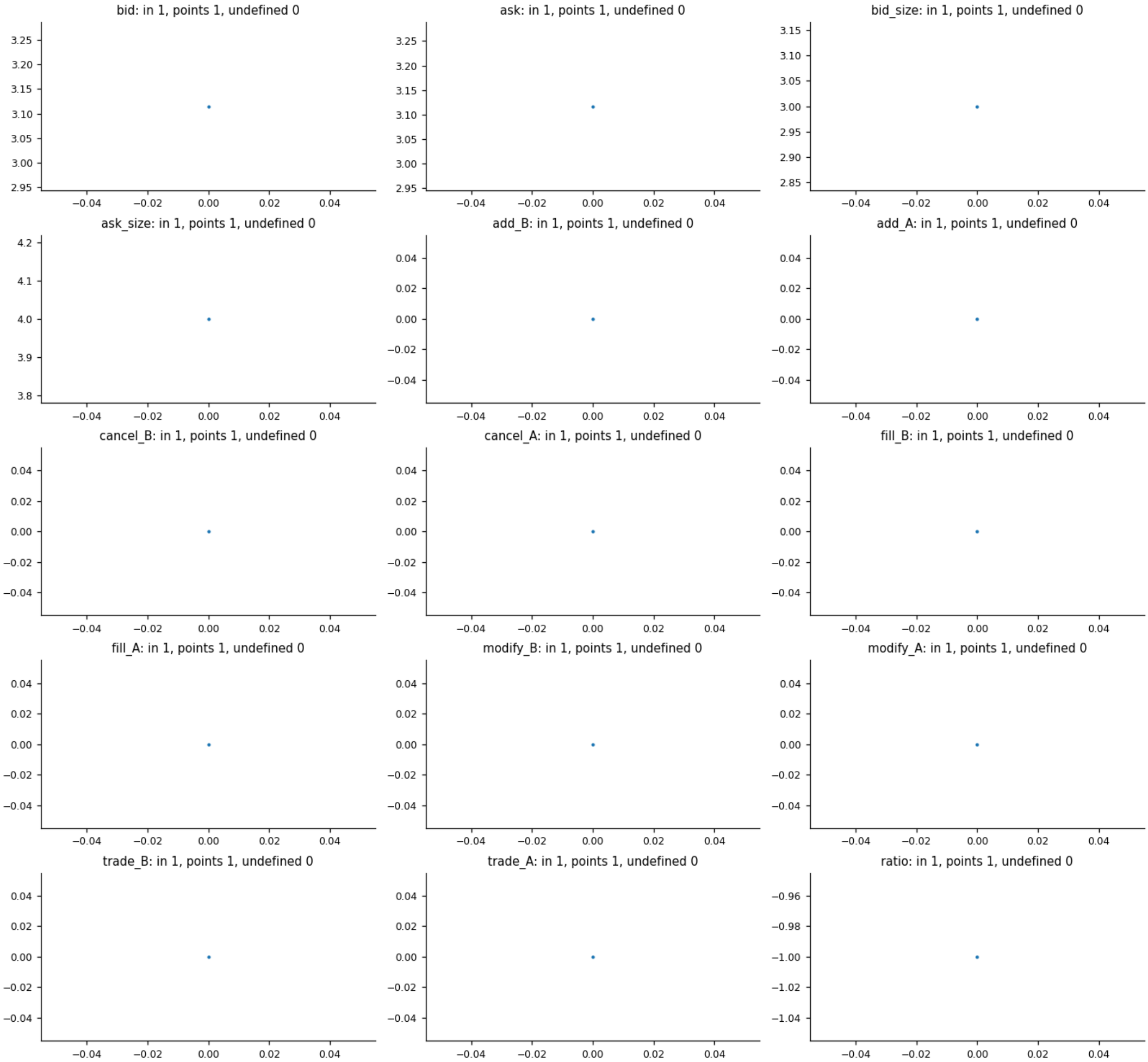
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



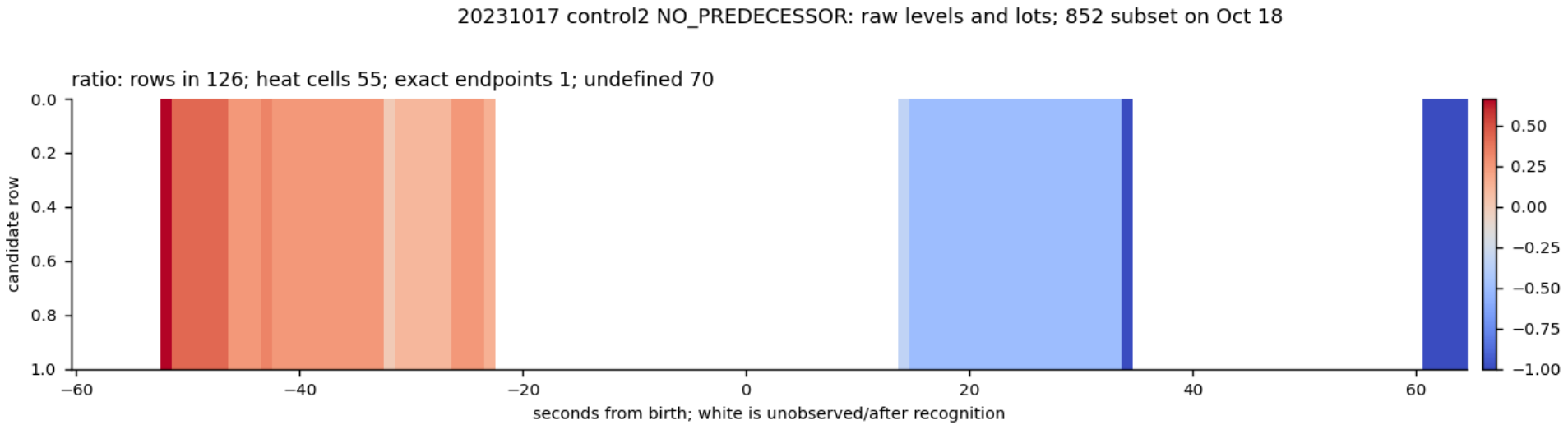

20231017 control2 NO_PREDECESSOR: raw levels and lots; 852 subset on Oct 18
add_B: rows in 126; heat cells 125; exact endpoints 1; undefined 0
add_A: rows in 126; heat cells 125; exact endpoints 1; undefined 0
cancel_B: rows in 126; heat cells 125; exact endpoints 1; undefined 0
cancel_A: rows in 126; heat cells 125; exact endpoints 1; undefined 0
fill_B: rows in 126; heat cells 125; exact endpoints 1; undefined 0
fill_A: rows in 126; heat cells 125; exact endpoints 1; undefined 0
modify_B: rows in 126; heat cells 125; exact endpoints 1; undefined 0
modify_A: rows in 126; heat cells 125; exact endpoints 1; undefined 0
trade_B: rows in 126; heat cells 125; exact endpoints 1; undefined 0
trade_A: rows in 126; heat cells 125; exact endpoints 1; undefined 0
candidate row
seconds from birth; white is unobserved/after recognition

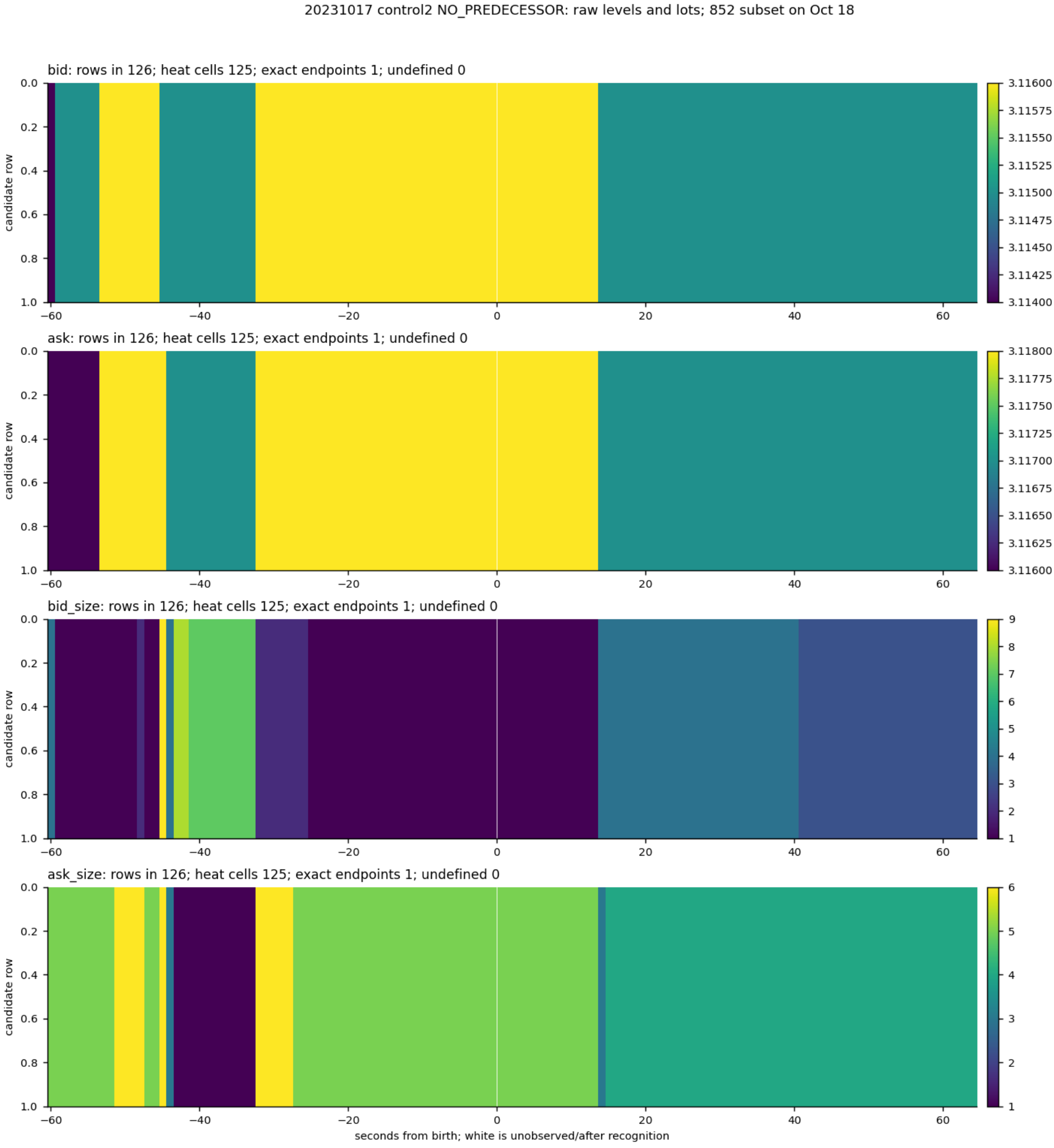
20231017-control2-NO_PREDECESSOR-endpoints.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

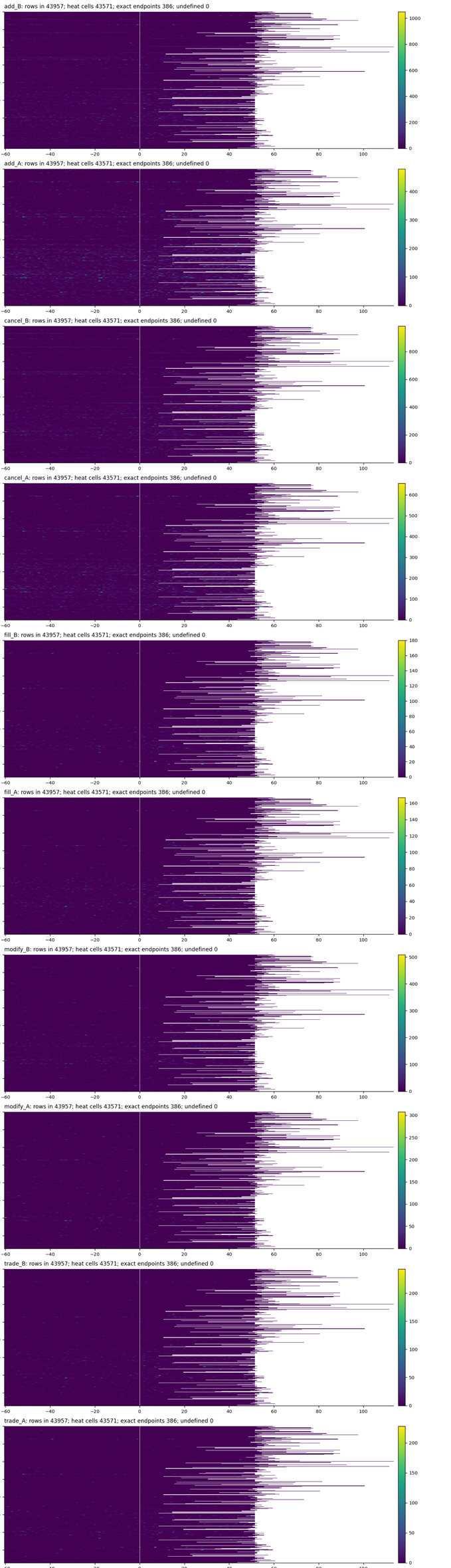


Original published render, unchanged. Individual source tables and figure reconciliation remain linked.



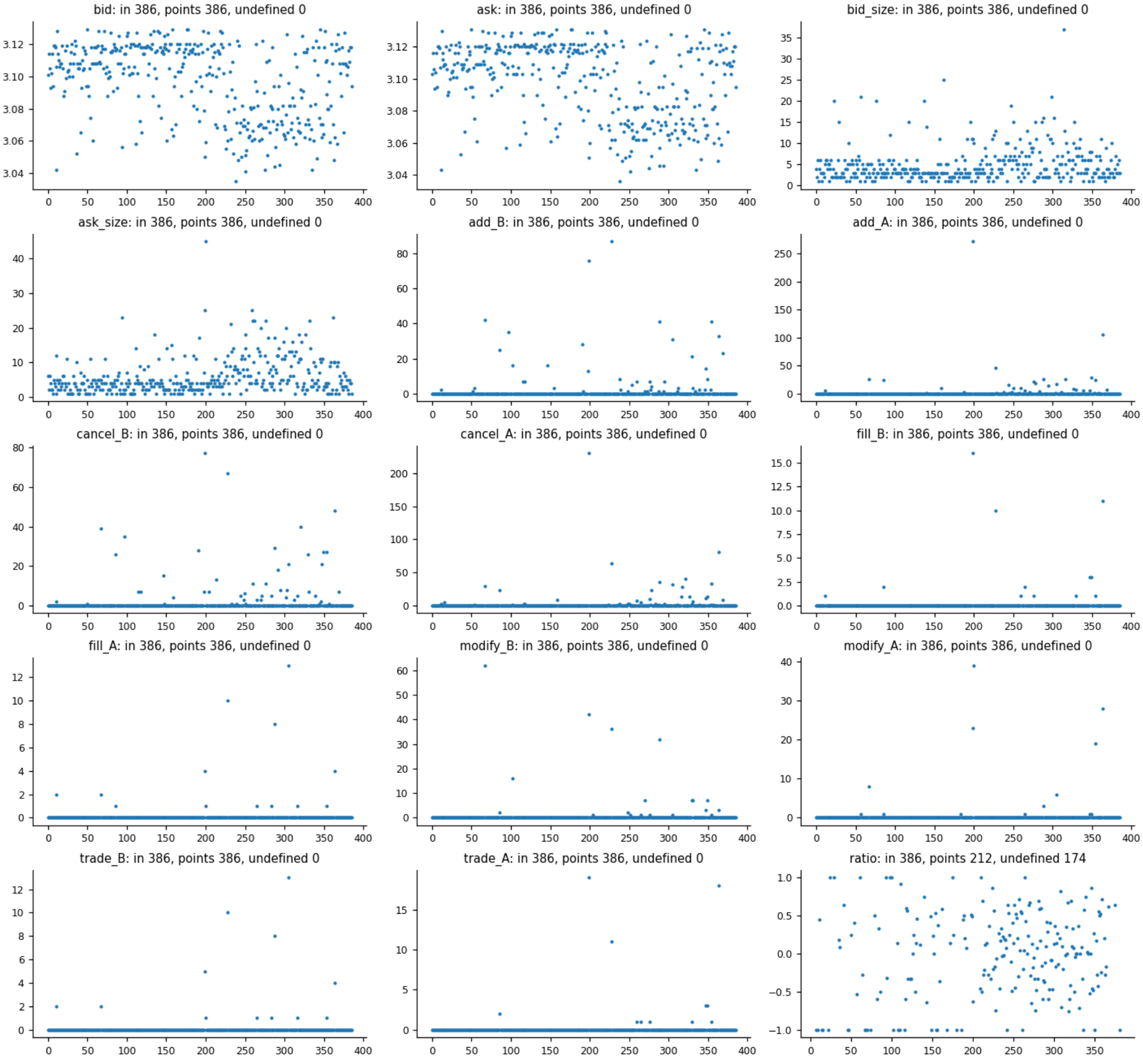

20231017 control2 NO_PREDECESSOR: raw levels and lots; 852 subset on Oct 18
bid: rows in 126; heat cells 125; exact endpoints 1; undefined 0
ask: rows in 126; heat cells 125; exact endpoints 1; undefined 0
bid_size: rows in 126; heat cells 125; exact endpoints 1; undefined 0
ask_size: rows in 126; heat cells 125; exact endpoints 1; undefined 0
candidate row
seconds from birth; white is unobserved/after recognition

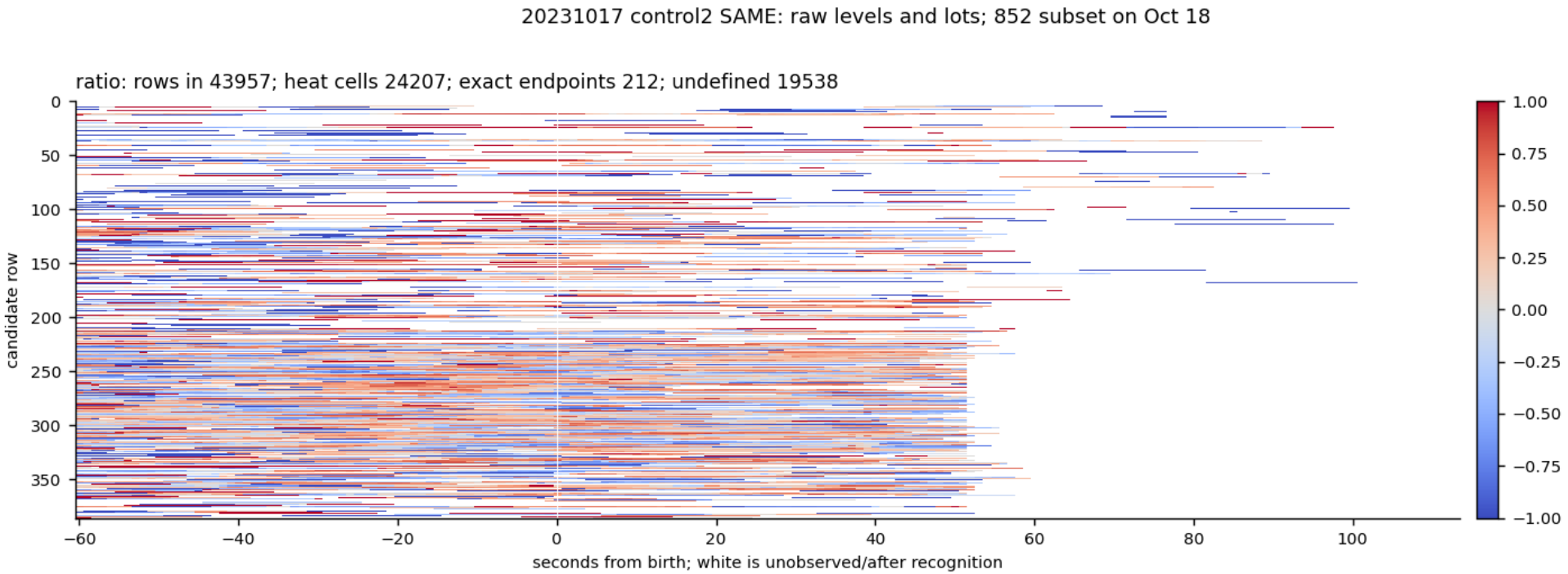
20231017 control2 SAME: raw levels and lots; 852 subset on Oct 18



20231017-control2-SAME-endpoints.png

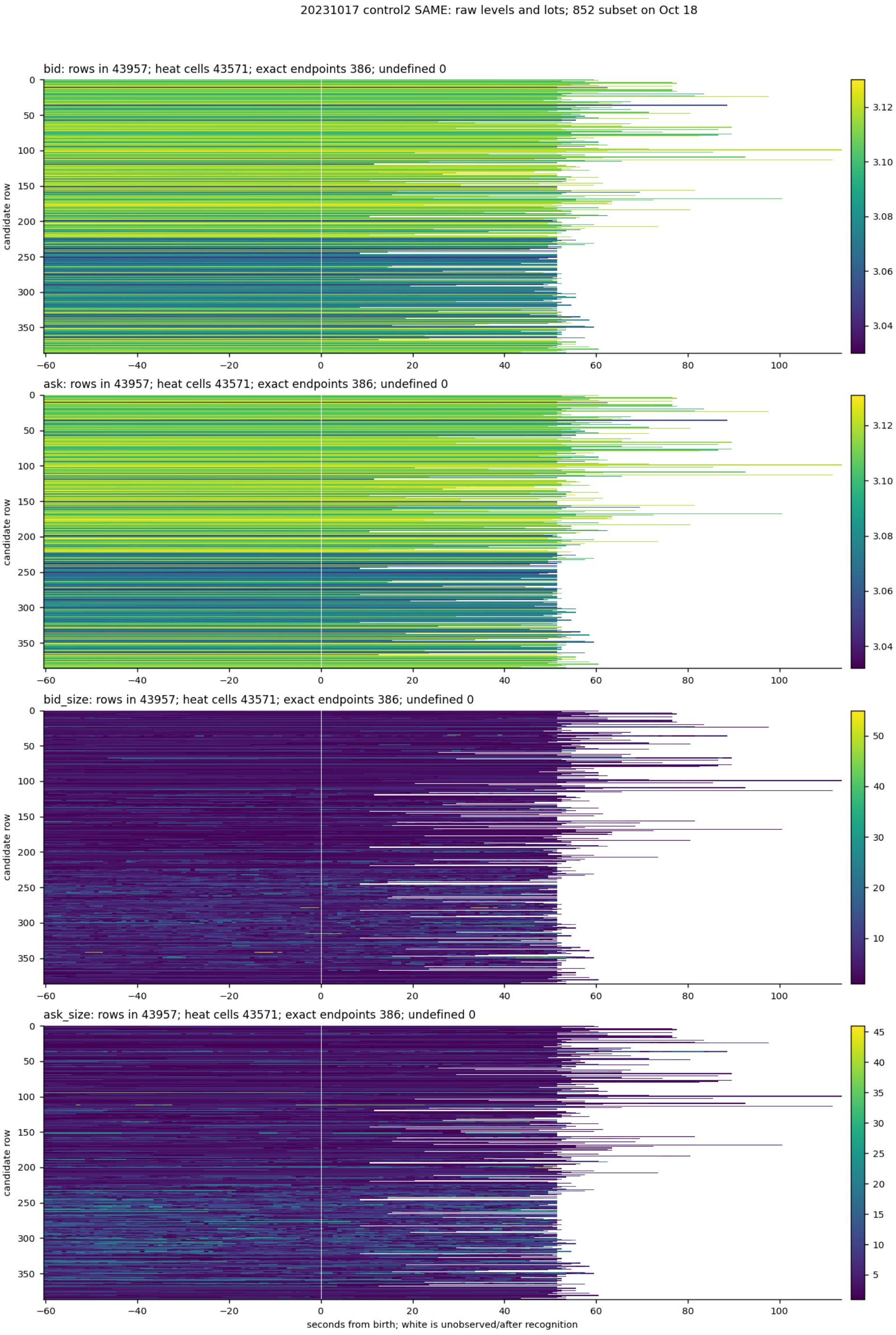
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

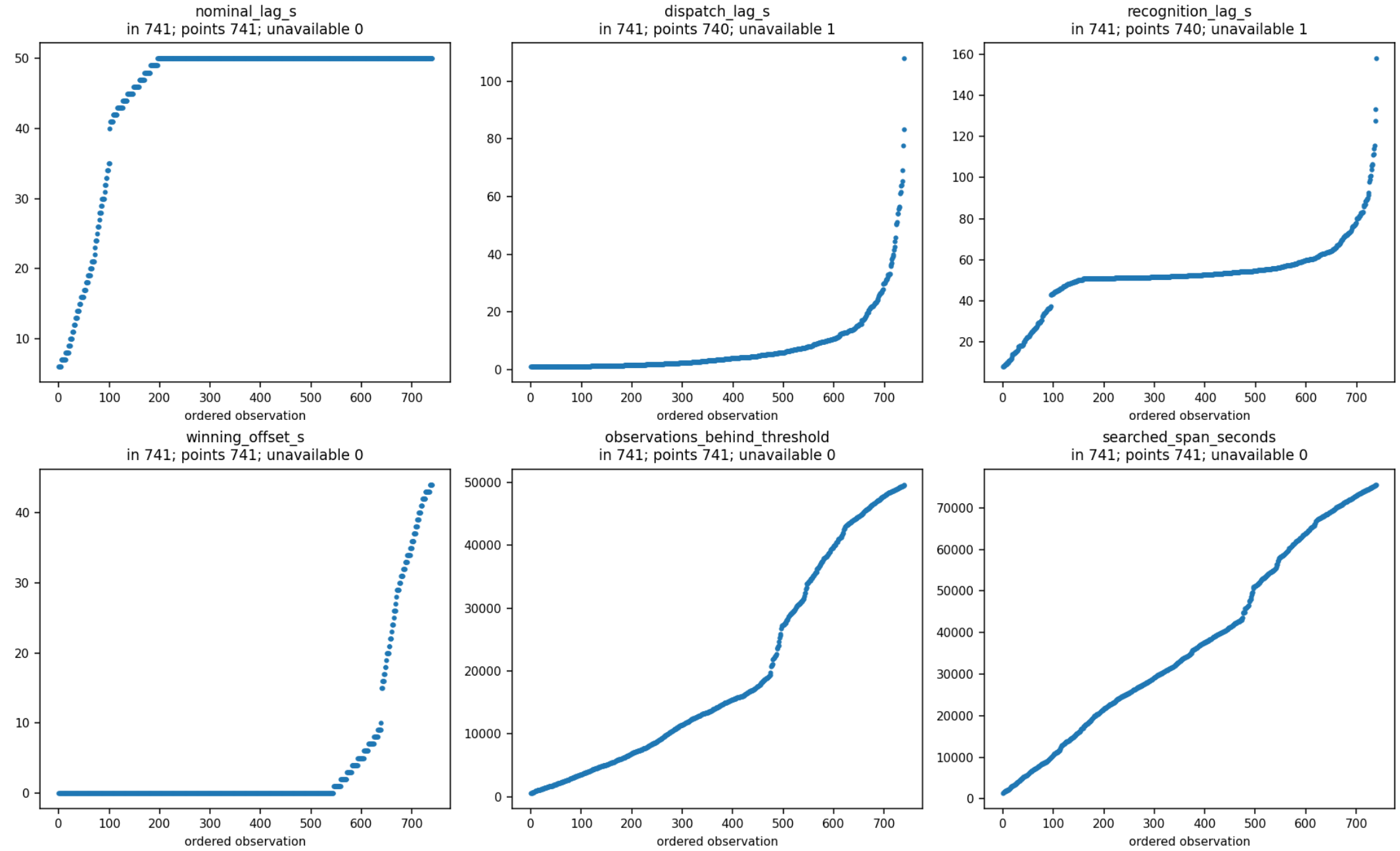


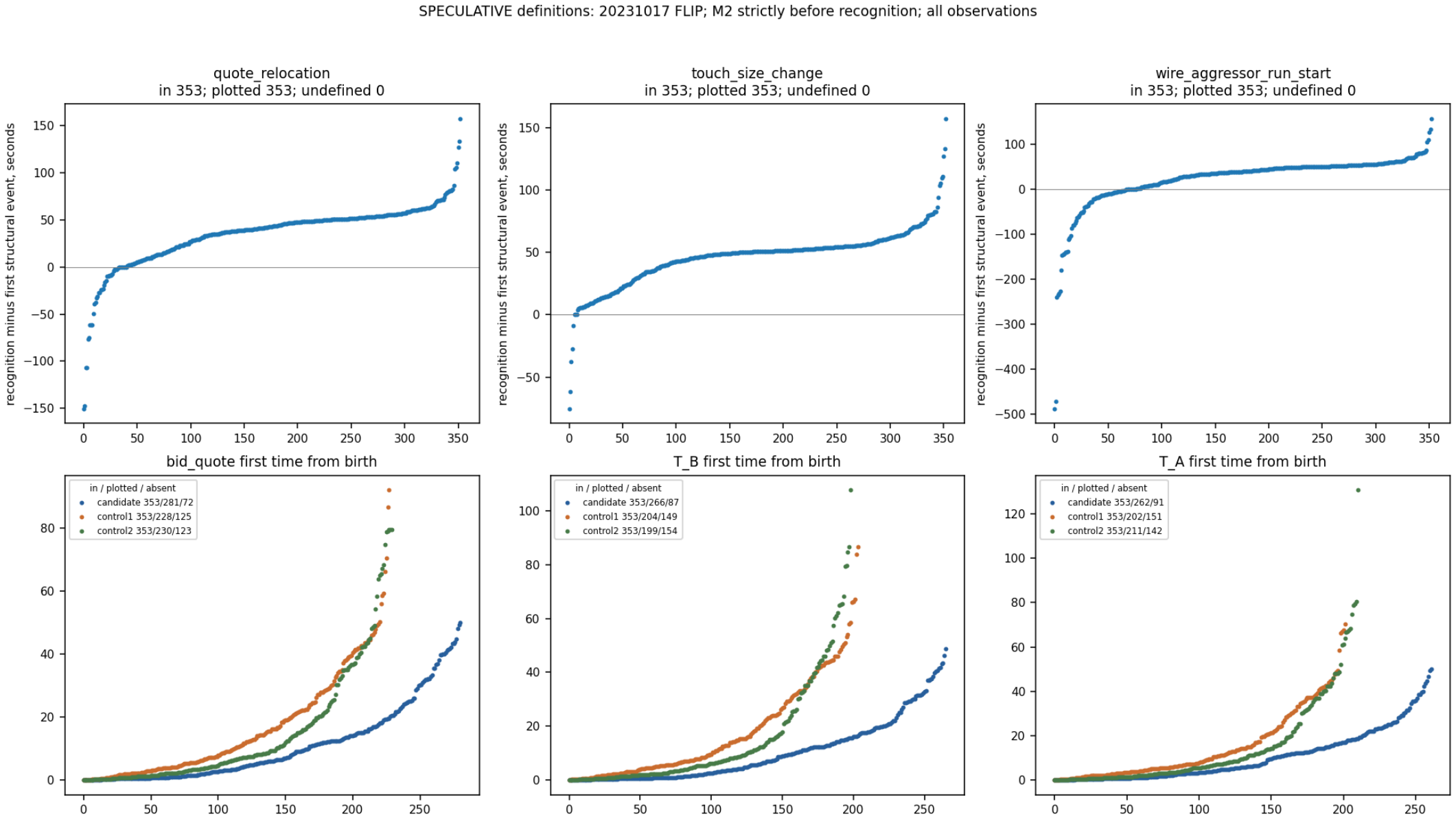

20231017 control2 SAME: raw levels and lots; 852 subset on Oct 18
ratio: rows in 43957; heat cells 24207; exact endpoints 212; undefined 19538
candidate row
0
50
100
150
200
250
300
350
−60
−40
−20
0
20
40
60
80
100
seconds from birth; white is unobserved/after recognition
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

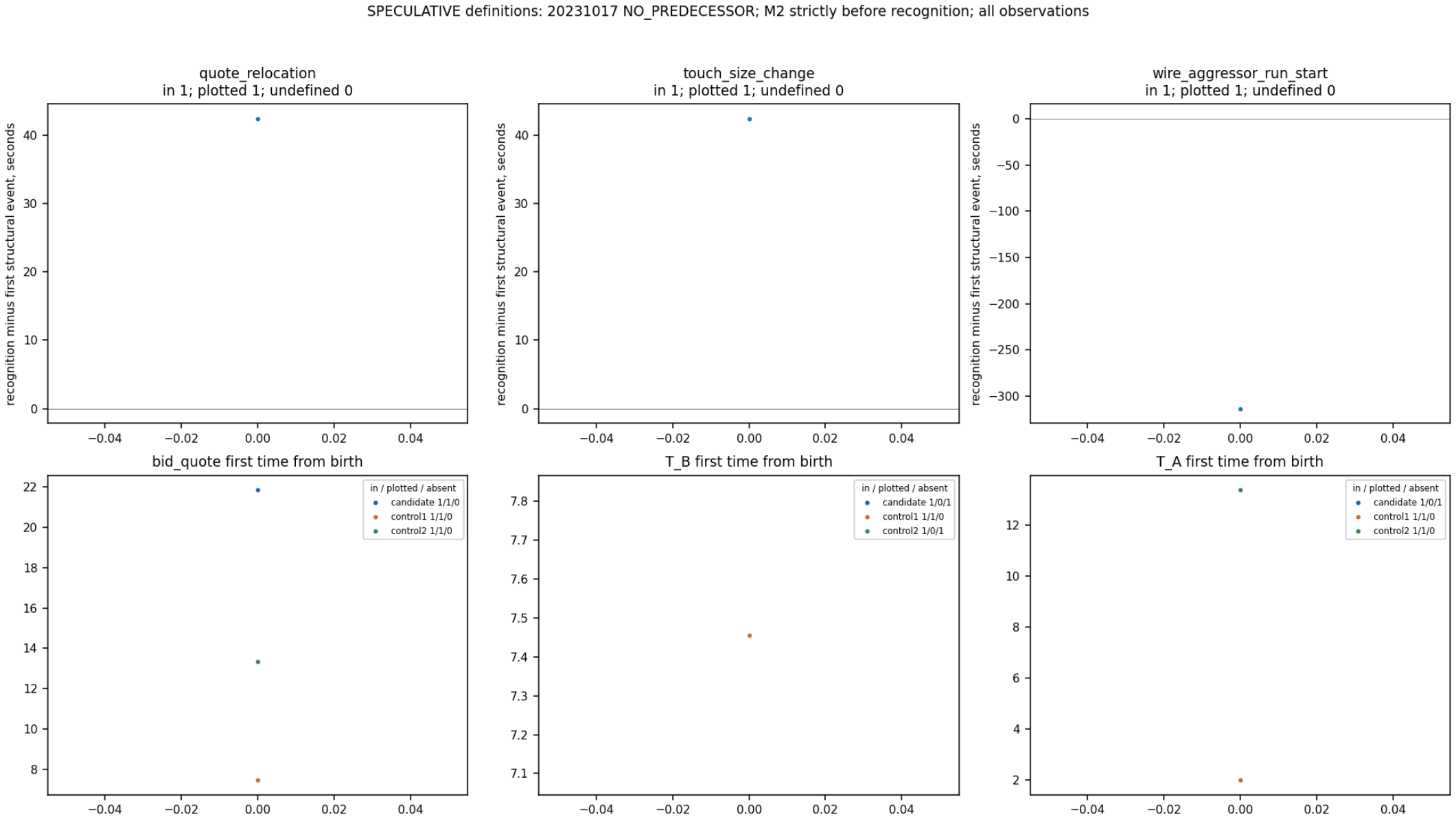
20231017-control2-SAME-quotes.png

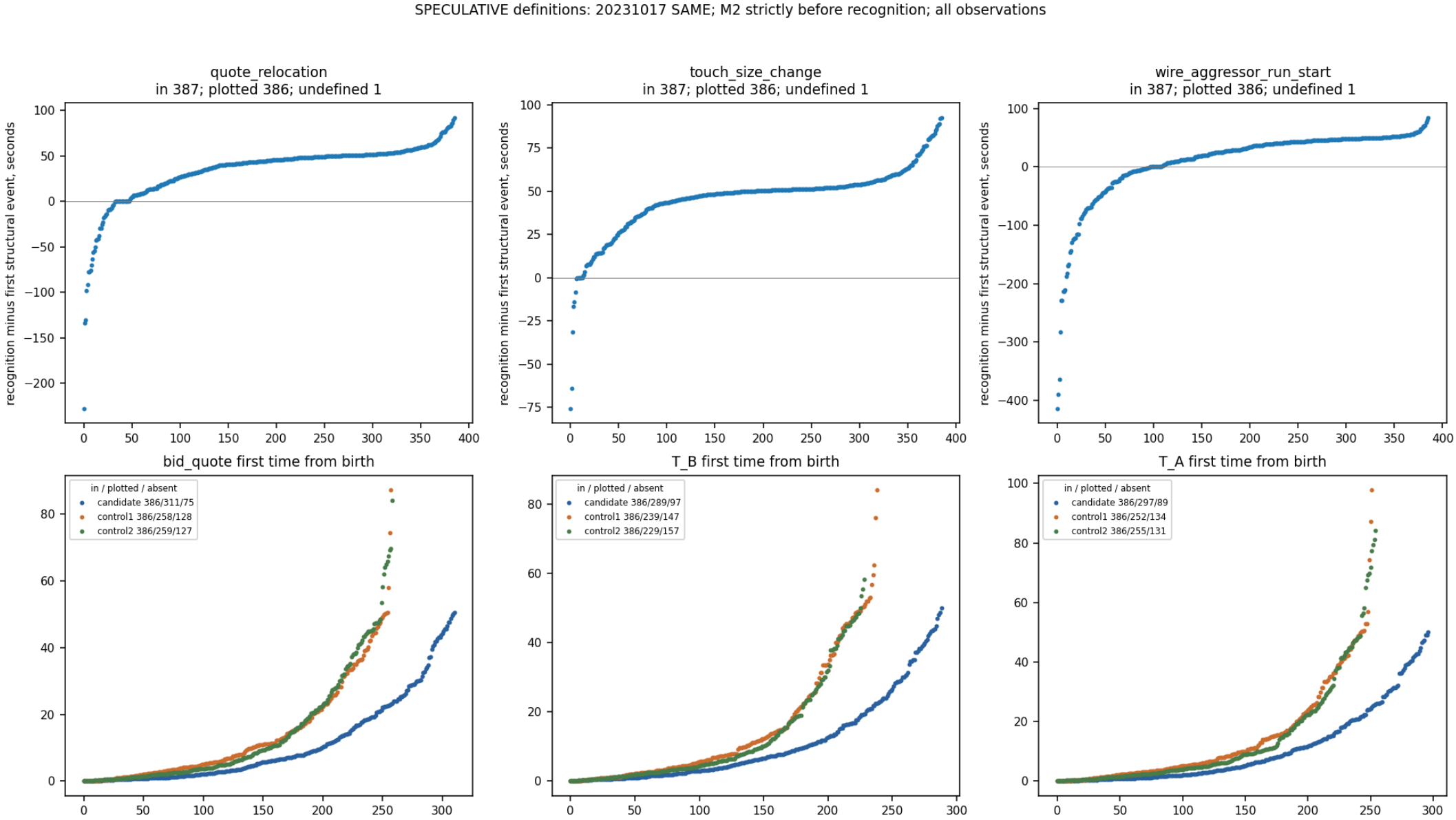
Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

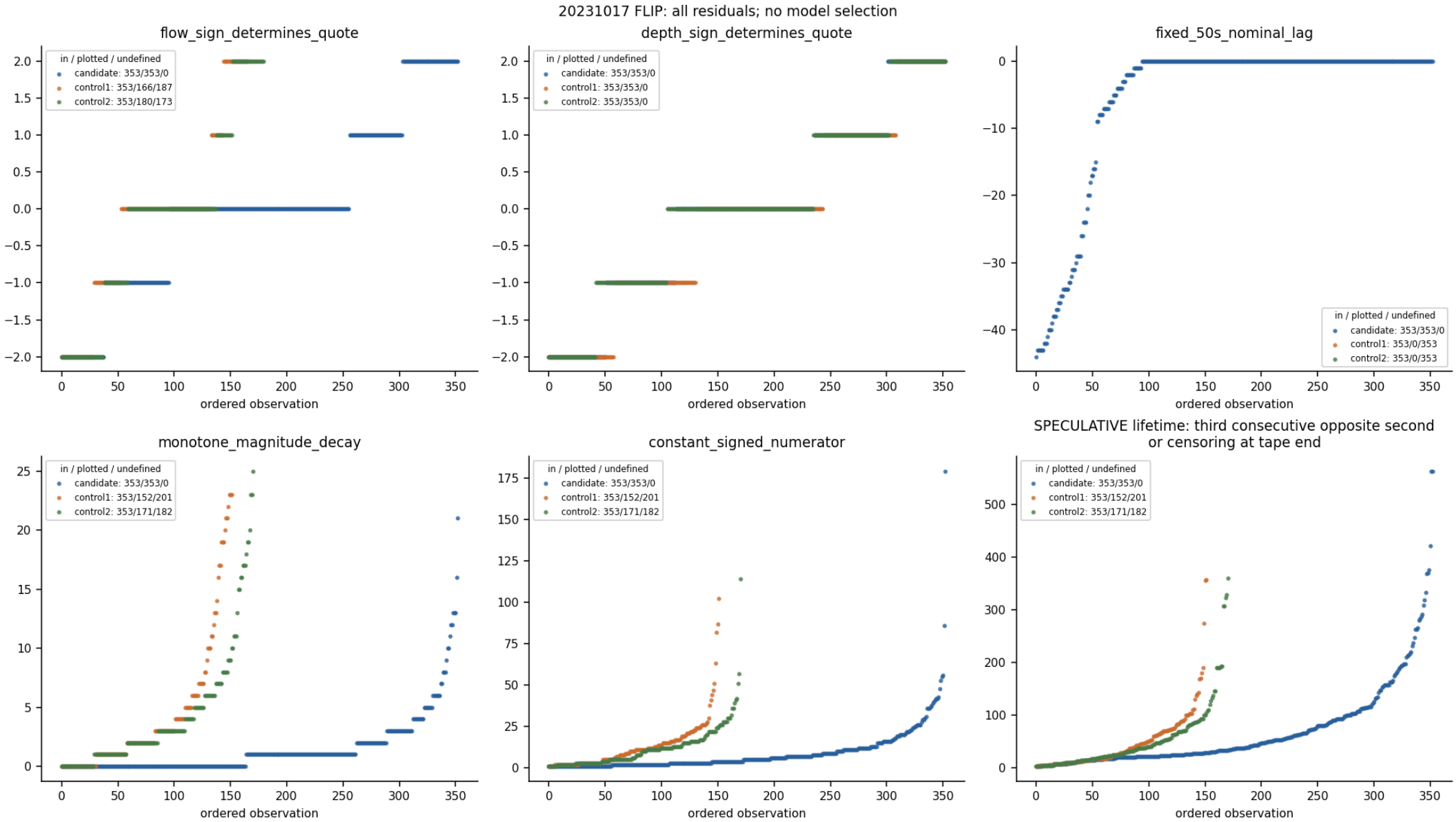


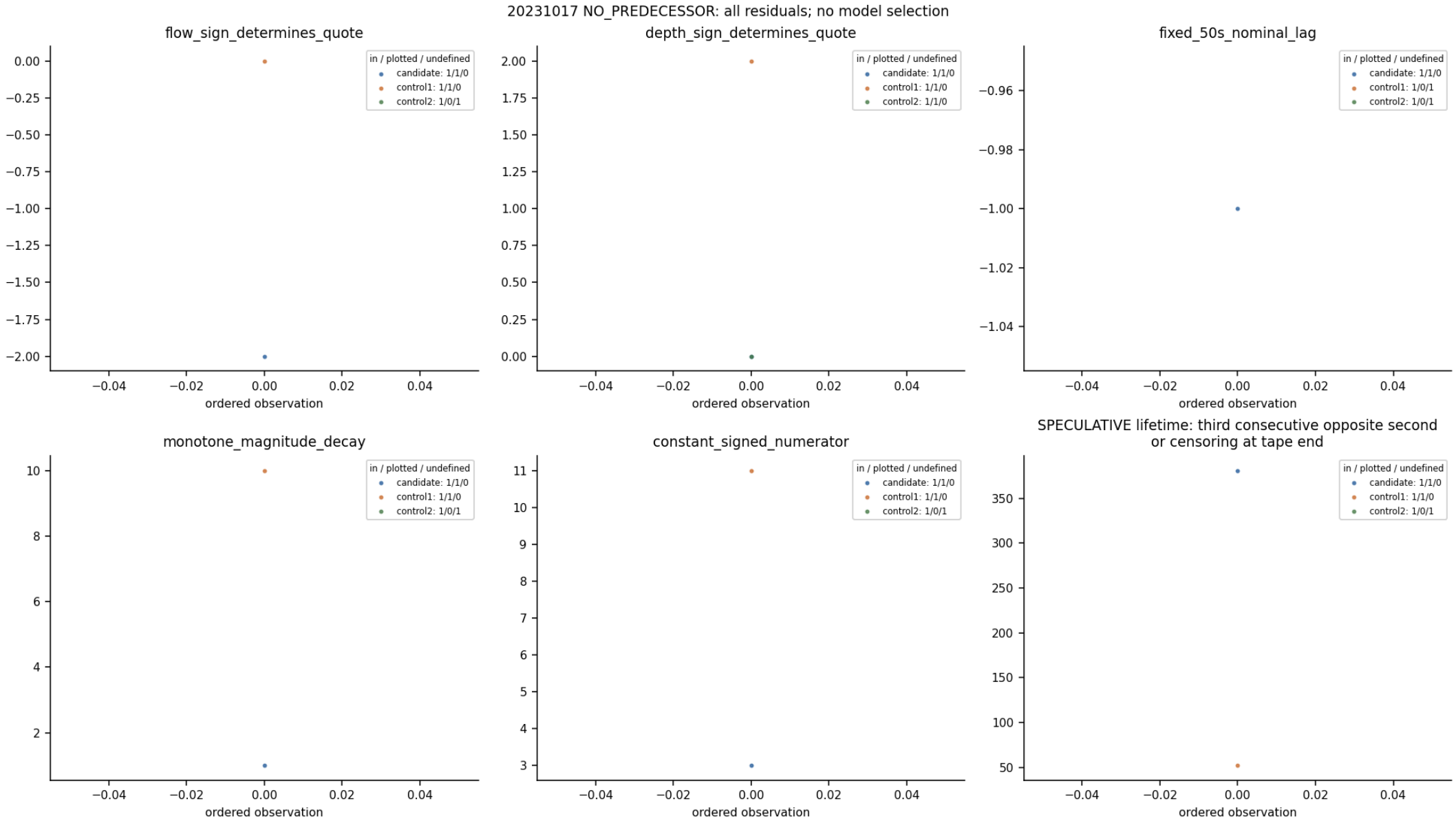

nominal_lag_s
in 741; points 741; unavailable 0
dispatch_lag_s
in 741; points 740; unavailable 1
recognition_lag_s
in 741; points 740; unavailable 1
winning_offset_s
in 741; points 741; unavailable 0
observations_behind_threshold
in 741; points 741; unavailable 0
searched_span_seconds
in 741; points 741; unavailable 0
ordered observation

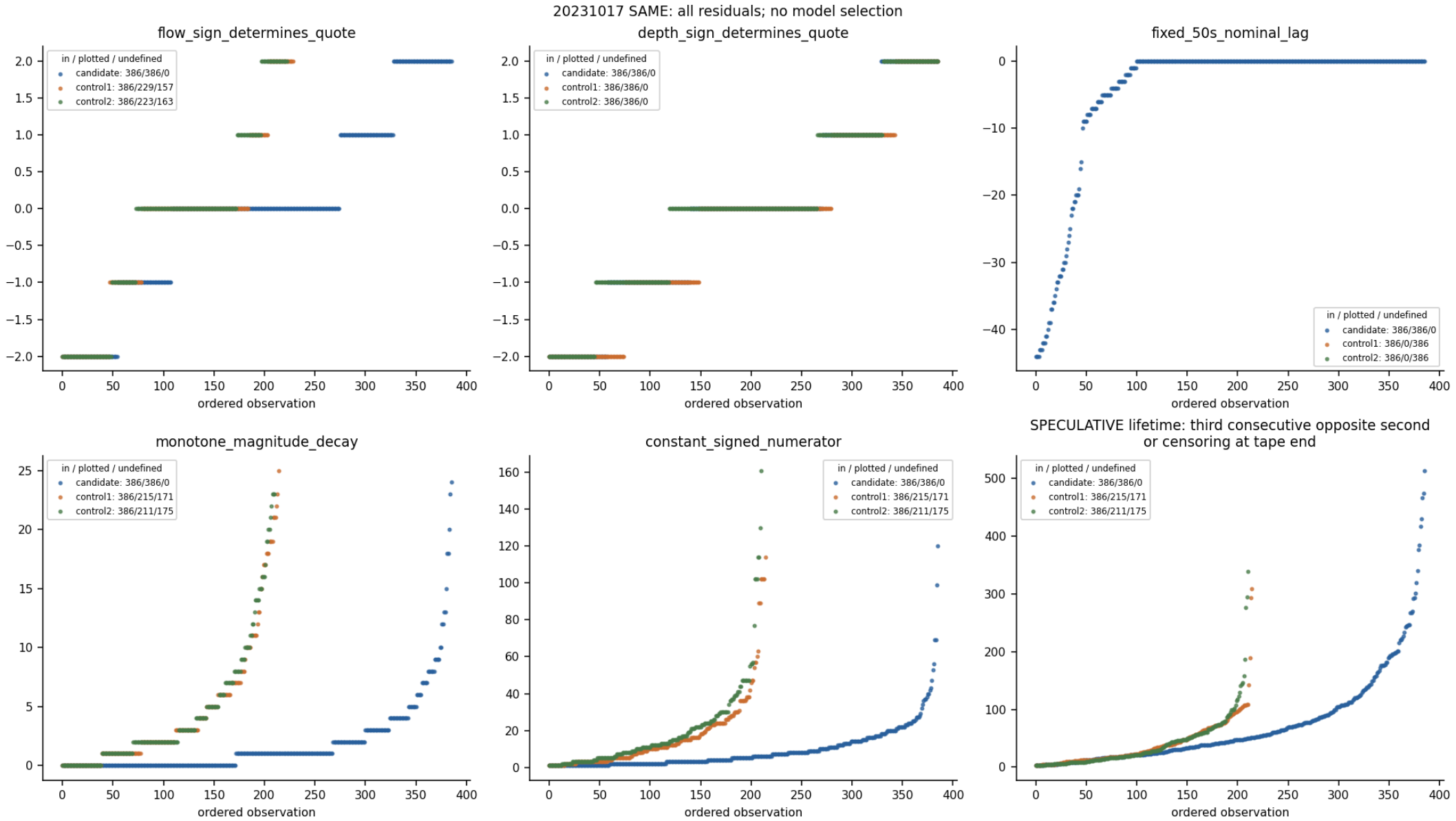

SPECULATIVE definitions: 20231017 FLIP; M2 strictly before recognition; all observations
quote_relocation
in 353; plotted 353; undefined 0
touch_size_change
in 353; plotted 353; undefined 0
wire_aggressor_run_start
in 353; plotted 353; undefined 0
recognition minus first structural event, seconds
bid_quote first time from birth
T_B first time from birth
T_A first time from birth
in / plotted / absent
candidate 353/281/72
control1 353/228/125
control2 353/230/123
candidate 353/266/87
control1 353/204/149
control2 353/199/154
candidate 353/262/91
control1 353/202/151
control2 353/211/142

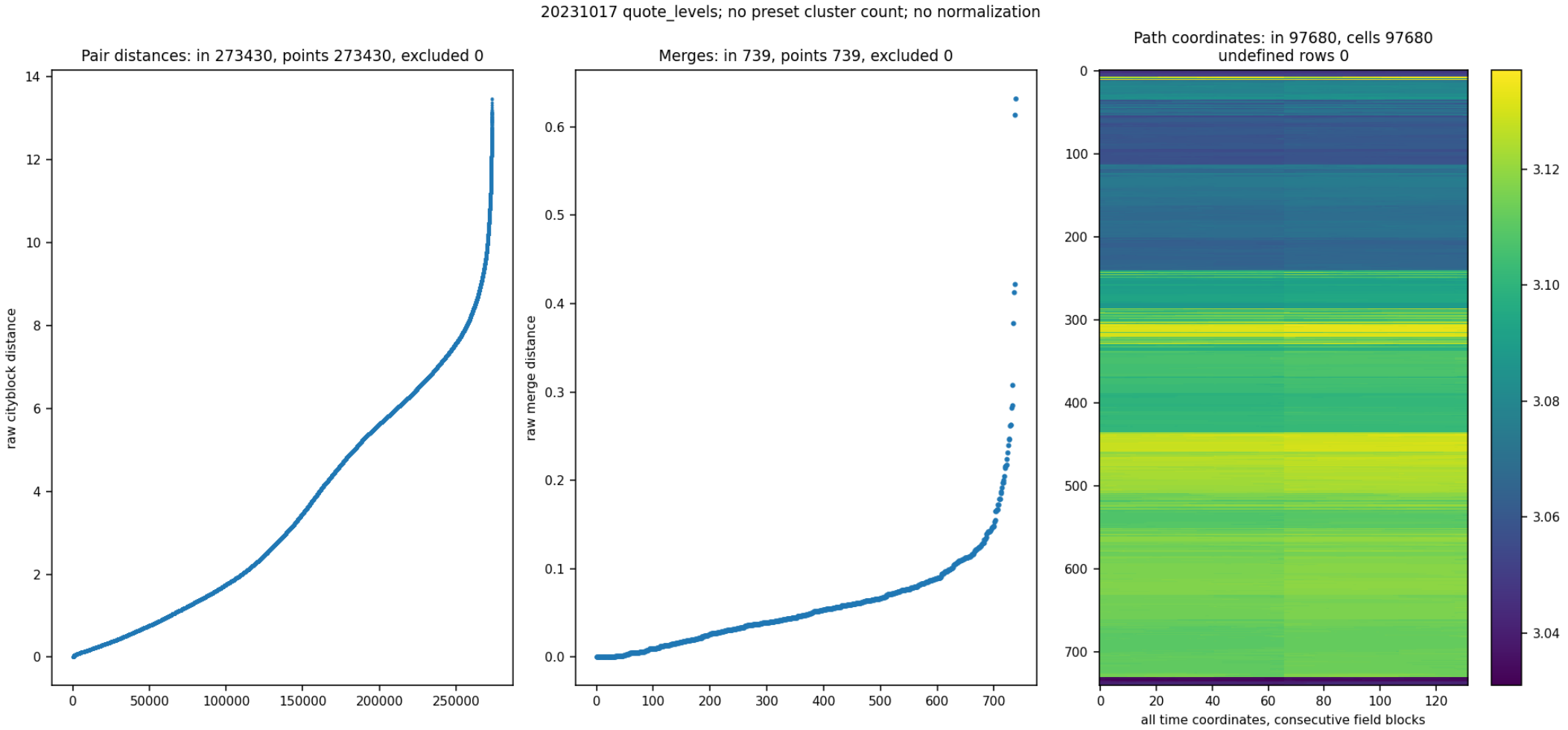

SPECULATIVE definitions: 20231017 NO_PREDECESSOR; M2 strictly before recognition; all observations
quote_relocation
in 1; plotted 1; undefined 0
touch_size_change
in 1; plotted 1; undefined 0
wire_aggressor_run_start
in 1; plotted 1; undefined 0
recognition minus first structural event, seconds
bid_quote first time from birth
T_B first time from birth
T_A first time from birth
in / plotted / absent
candidate 1/1/0
control1 1/1/0
control2 1/1/0
candidate 1/0/1
control1 1/1/0
control2 1/0/1
candidate 1/0/1
control1 1/1/0
control2 1/1/0


SPECULATIVE definitions: 20231017 SAME; M2 strictly before recognition; all observations
quote_relocation
in 387; plotted 386; undefined 1
touch_size_change
in 387; plotted 386; undefined 1
wire_aggressor_run_start
in 387; plotted 386; undefined 1
recognition minus first structural event, seconds
bid_quote first time from birth
T_B first time from birth
T_A first time from birth
in / plotted / absent
candidate 386/311/75
control1 386/258/128
control2 386/259/127
candidate 386/289/97
control1 386/239/147
control2 386/229/157
candidate 386/297/89
control1 386/252/134
control2 386/255/131


20231017 FLIP: all residuals; no model selection
flow_sign_determines_quote
depth_sign_determines_quote
fixed_50s_nominal_lag
monotone_magnitude_decay
constant_signed_numerator
SPECULATIVE lifetime: third consecutive opposite second or censoring at tape end
in / plotted / undefined
candidate: 353/353/0
control1: 353/166/187
control2: 353/180/173
control1: 353/353/0
control2: 353/353/0
control1: 353/0/353
control2: 353/0/353
control1: 353/152/201
control2: 353/171/182
ordered observation

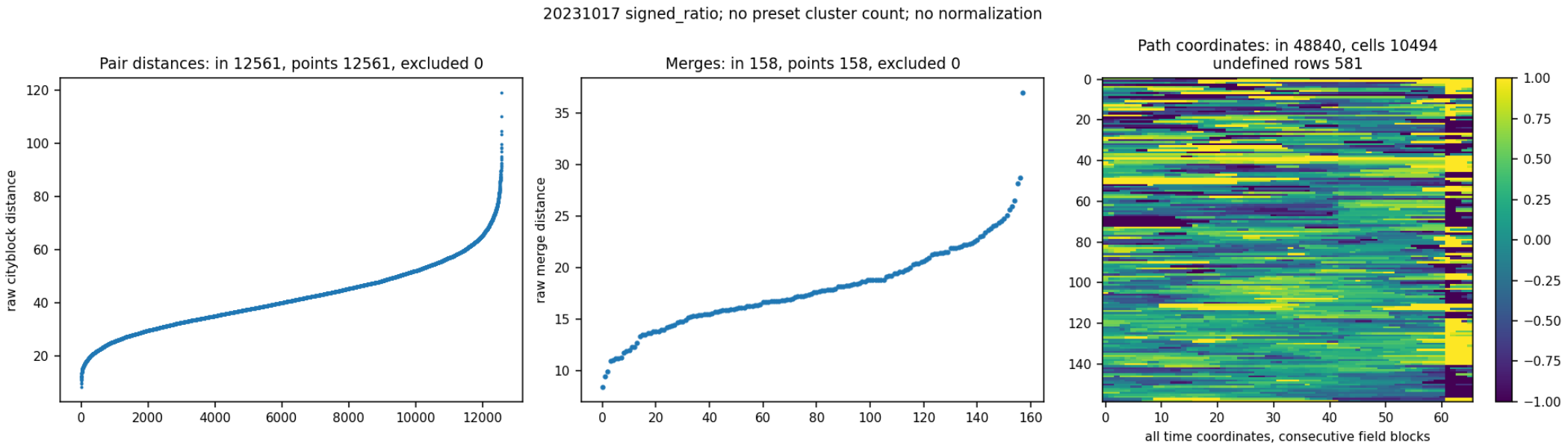

20231017 NO_PREDECESSOR: all residuals; no model selection
flow_sign_determines_quote
depth_sign_determines_quote
fixed_50s_nominal_lag
monotone_magnitude_decay
constant_signed_numerator
SPECULATIVE lifetime: third consecutive opposite second or censoring at tape end
in / plotted / undefined
candidate: 1/1/0
control1: 1/1/0
control2: 1/0/1
control1: 1/0/1
ordered observation

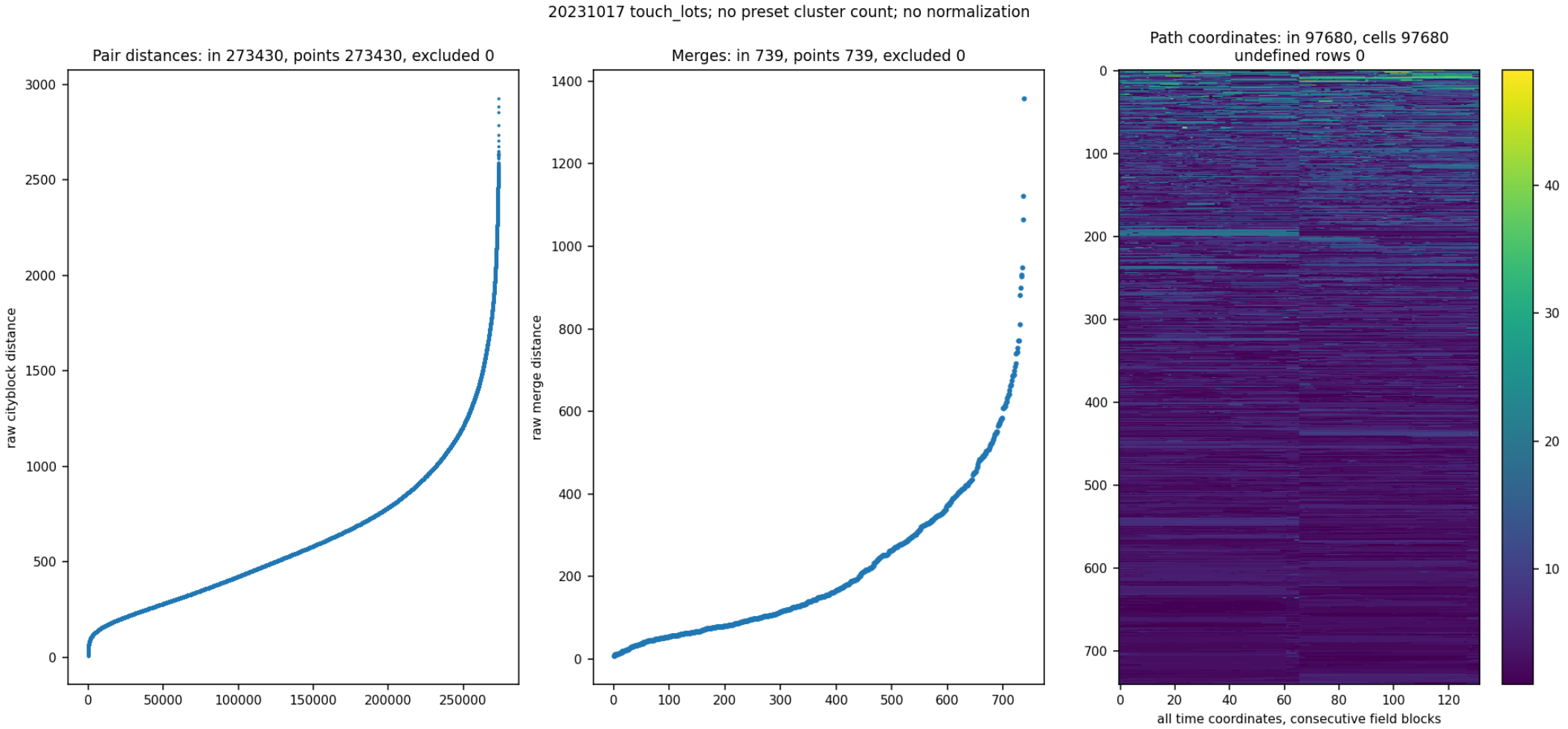

20231017 SAME: all residuals; no model selection
flow_sign_determines_quote
depth_sign_determines_quote
fixed_50s_nominal_lag
monotone_magnitude_decay
constant_signed_numerator
SPECULATIVE lifetime: third consecutive opposite second or censoring at tape end
in / plotted / undefined
candidate: 386/386/0
control1: 386/229/157
control2: 386/223/163
control1: 386/386/0
control2: 386/386/0
control1: 386/0/386
control2: 386/0/386
control1: 386/215/171
control2: 386/211/175
ordered observation


20231017 quote_levels; no preset cluster count; no normalization
Pair distances: in 273430, points 273430, excluded 0
raw cityblock distance
Merges: in 739, points 739, excluded 0
raw merge distance
Path coordinates: in 97680, cells 97680
undefined rows 0
all time coordinates, consecutive field blocks

20231017-structure-signed_ratio.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.




20231017 touch_lots; no preset cluster count; no normalization
Pair distances: in 273430, points 273430, excluded 0
raw cityblock distance
Merges: in 739, points 739, excluded 0
raw merge distance
Path coordinates: in 97680, cells 97680
undefined rows 0
all time coordinates, consecutive field blocks

20231017-structure-wire_trade_lots.png

Original published render, unchanged. Individual source tables and figure reconciliation remain linked.

